A ZERO-FREE REGION FOR THE RIEMANN ZETA FUNCTION IN THE HALF-PLANE $\Re s \geq 0.6$

JONATHAN WASHBURN AND AMIR RAHNAMAI BARGHI

Abstract. We establish a zero-free region for the Riemann zeta function: $\zeta(s) \neq 0$ for $\Re s \geq 0.6$. On $\Omega = \{\Re s > \frac{1}{2}\}$ we introduce an arithmetic ratio $\mathcal{J}(s)$ and the associated Cayley transform $\Theta(s) = (2\mathcal{J}(s) - 1)/(2\mathcal{J}(s) + 1)$. A boundary wedge condition (P^+) implies that Θ is Schur on Ω, which in turn yields a half-plane Schur/Herglotz removability argument excluding poles of $\mathcal{J}$ throughout $\{\Re s \geq 0.6\}$. Since any pole of $\mathcal{J}$ in Ω forces a zero of ζ, the pole exclusion gives the stated zero-free region. Supplementary rigorous ball-arithmetic certificates are included as ancillary material for independent reproducibility checks on representative low-height rectangles.

1. Introduction

The Riemann zeta function

$$\zeta(s) \;=\; \sum_{n\geq 1} \frac{1}{n^s}, \qquad \Re s > 1,$$

extends meromorphically to $\mathbb{C}$ with a simple pole at $s = 1$ and satisfies a functional equation after completion. Its nontrivial zeros govern the finest fluctuations in the distribution of prime numbers, and the Riemann Hypothesis (RH) asserts that all such zeros lie on the critical line $\Re s = \frac{1}{2}$; see [1, 3] for background.

This paper isolates an unconditional, fixed half-plane statement in the direction of RH. Unlike the classical zero-free regions near $\Re s = 1$ (which are asymptotic in height), the result here is a *uniform* half-plane exclusion at $\Re s \geq 0.6$.

Theorem 1 (far-field zero-freeness)**.** *The Riemann zeta function has no zeros in the region* $\{\, s \in \mathbb{C} :\ \Re s \geq 0.6 \,\}$.
Proof. See §4.1.

Scope and supplementary computation. The proof of Theorem 1 is analytic and self-contained in the present text, with the boundary wedge certificate (P+) proved in §§A.3–A.5 of Appendix A. The shipped computational artifacts provide independently checkable ball-arithmetic cross-checks on representative low-height rectangles (Table 1) together with a verification protocol (Appendix B). They are included for transparency and reproducibility, but are not logically required for the all-heights implication chain in the proof.

The Appendix records the analytic function theory inputs used in the proof of (P+) (boundary traces, factorization, and the phase–velocity identity).

Strategy: Schur pinching via a Cayley field. We work on the right half-plane $\Omega = \{\, \Re s > \frac{1}{2} \,\}$. In Section 2 we define an arithmetic ratio $\mathcal{J}$ (in the default *raw ζ-gauge*) with the following two structural properties:

- **(normalization at $+\infty$)** $\mathcal{J}(\sigma + it) \to 1$ as $\sigma \to +\infty$, hence $\Theta(\sigma + it) \to \frac{1}{3}$ (Remark 3);

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Recognition Physics Research Institute, Austin, TX, USA; .
Correspondence: `arahnama@gmail.com`.
Recognition Physics Research Institute, Austin, TX, USA; .
Correspondence: arahnamab@gmail.com.

- **(normalization)** $\det_2(I - A(s))$ is holomorphic and nonvanishing on Ω, so any zero of ζ in Ω produces a pole of $\mathcal{J}$ (Lemma 4).

We then pass to the Cayley transform

$$\Theta(s) \;:=\; \frac{2\mathcal{J}(s) - 1}{2\mathcal{J}(s) + 1}.$$

The analytic mechanism is a *Schur/Herglotz pinch* proved in Section 3: if Θ is Schur on a domain (i.e. $|\Theta| \le 1$) and not identically 1, then boundedness forces removability of any isolated singularity and prevents poles of $\mathcal{J}$. Since $\Theta(\sigma + it) \to \frac{1}{3}$ as $\sigma \to +\infty$, the degenerate possibility $\Theta \equiv 1$ is excluded on the half-planes relevant here. Therefore, to prove Theorem 1 it suffices to certify a Schur bound for the default Cayley field Θ_{raw} on some open half-plane $\{\, \Re s > 0.6 - \varepsilon \,\}$.

Inputs (what is rigorously checked). The logical implication of Theorem 1 rests on a boundary certificate: we establish a boundary wedge (P+) (Lebesgue-a.e.) for the boundary phase of $\mathcal{J}$ on $\Re s = \frac{1}{2}$, which implies that $2\mathcal{J}$ is Herglotz and Θ is Schur on $\Omega \setminus Z(\zeta)$. The Schur/Herglotz pinch mechanism then excludes poles of $\mathcal{J}$ on $\{\, \Re s \ge 0.6 \,\}$ and hence excludes zeros of ζ there.

Supplementary computational cross-checks (not used in the proof). The handoff bundle also contains rigorous ball-arithmetic rectangle checks and finite Pick artifacts on low-height regions; these are included as independent numerical corroboration but are not used in the all-heights proof.

Reproducibility and verification materials. The handoff bundle (and repository) includes: (i) the verifier script based on ARB ball arithmetic (`python-flint`), and (ii) the JSON artifacts that record the maxima, spectral separation bounds, and denominator checks for the supplementary certificates listed in Table 1. The included `MANIFEST.md` and Appendix B provide a verification manifest mapping certificates to exact commands and expected output fields.

Place in a series. This paper is designed to stand alone as an unconditional zero-free region. Two companion papers (not required for Theorem 1) treat: (a) effective near-field energy barriers and Carleson budgets, and (b) a cutoff principle yielding conditional closure of RH.

The remainder of the paper defines the arithmetic ratio $\mathcal{J}$ and Cayley field Θ, proves the Schur pinch mechanism, and then discharges the Schur bound via the hybrid certification outlined above.

2. Definitions and main objects

This section defines the analytic objects used throughout the proof and records the basic relationships between zeros of ζ and the bounded-real (Schur/Herglotz) structure employed later.

The completed zeta function and the far half-plane. Let $\zeta(s)$ denote the Riemann zeta function. We write $\xi(s)$ for the completed zeta function

$$\xi(s) \;:=\; \tfrac{1}{2}\, s(s-1)\, \pi^{-s/2}\Gamma(s/2)\, \zeta(s),$$

which is entire and satisfies the functional equation $\xi(s) = \xi(1-s)$; see [3]. In this paper, when we say "zero" we mean a zero of ζ (equivalently of ξ away from the canceled singularities at $s = 0, 1$) lying in the half-plane

$$\Omega \;:=\; \{\, s \in \mathbb{C} :\; \Re s > \tfrac{1}{2} \,\}.$$

Theorem 1 concerns the fixed far region $\{\, \Re s \ge 0.6 \,\} \subset \Omega$.

The prime-diagonal operator and the regularized determinant. Let $\mathcal{P}$ denote the set of primes and write $\ell^2(\mathcal{P})$ for the Hilbert space with orthonormal basis $\{e_p\}_{p\in\mathcal{P}}$. For $s \in \mathbb{C}$ define the prime-diagonal operator

$$A(s) : \ell^2(\mathcal{P}) \to \ell^2(\mathcal{P}), \qquad A(s)e_p := p^{-s}e_p.$$

For $\Re s > 1/2$,

$$\|A(s)\|_{\mathrm{HS}}^2 = \sum_{p\in\mathcal{P}} |p^{-s}|^2 = \sum_{p\in\mathcal{P}} p^{-2\Re s} \le \sum_{n\ge 2} n^{-2\Re s} < \infty,$$

so $A(s)$ is Hilbert–Schmidt on Ω. In particular, the regularized determinant $\det_2(I - A(s))$ is well-defined and holomorphic on Ω (see [4, Ch. III] and [6, Ch. 9]).

Lemma 2 (Diagonal product formula for $\det_2$)**.** *Let T be a diagonal Hilbert–Schmidt operator on ℓ^2 with eigenvalues $\{\lambda_n\}$ satisfying $\sum_n |\lambda_n|^2 < \infty$. Then*

$$\det_2(I - T) \;=\; \prod_n (1 - \lambda_n)\, e^{\lambda_n},$$

where the product converges absolutely. In particular, $\det_2(I - T) = 0$ *iff* $\lambda_n = 1$ *for some* n.

Proof. This holds for the $\mathcal{S}_2$-regularized determinant; see [4, Ch. III] or [6, Ch. 9]. (We only use the diagonal case and the zero criterion $\lambda_n = 1$.) □

Applying Lemma 2 to $T = A(s)$ on Ω gives the explicit product

$$\det_2(I - A(s)) \;=\; \prod_{p\in\mathcal{P}} (1 - p^{-s})\, e^{p^{-s}}. \tag{1}$$

The region $\Omega \subset \{\Re s > 1/2\}$ lies away from $s = 0$, so the compensator $1/s$ introduces no pole on the working domain. The point $s = 1$ lies in Ω, but the factor $(s - 1)$ cancels the simple pole of ζ there. All holomorphy/pole assertions for $\mathcal{J}$ are made only on Ω, and poles are tracked relative to zeros of ζ in Ω.

Since $\Re s > 1/2$ implies $|p^{-s}| < 1$ for every prime p, each factor in (1) is nonzero. Hence $\det_2(I - A(s))$ is holomorphic and zero-free on Ω.

The arithmetic ratio $\mathcal{J}$ and the Cayley field Θ. Fix a domain $D \subset \Omega$. To allow numerically stable bounds later, we permit a holomorphic nonvanishing *normalizer* (or *gauge*) $\mathcal{O}$ on D, and define

$$\mathcal{J}(s) \;:=\; \frac{\det_2(I - A(s))}{\zeta(s)} \cdot \frac{s-1}{s} \cdot \frac{1}{\mathcal{O}(s)}, \qquad s \in D. \tag{2}$$

The factor $(s - 1)$ cancels the simple pole of ζ at $s = 1$; the factor $1/s$ plays no role on $D \subset \Omega$ (but is convenient in later normalization). Unless explicitly stated otherwise, we work in the *raw ζ-gauge* $\mathcal{O} \equiv 1$ and denote the resulting objects by $\mathcal{J}_{\mathrm{raw}}$ and Θ_{raw}; for readability we usually drop the subscript in this default gauge.

Remark 3 (Gauge changes and what they do *not* change). If $\mathcal{O}$ is holomorphic and nonvanishing on D, then multiplying by $\mathcal{O}^{-1}$ cannot introduce poles on D. Thus the pole set of $\mathcal{J}$ on D is independent of the choice of gauge. However, quantitative bounds (e.g. Schur bounds on Θ) are not gauge-invariant; when a nontrivial gauge is used for a bound, one also requires that $\mathcal{O}$ is holomorphic and nonvanishing on the domain. In the raw gauge $\mathcal{O} \equiv 1$ one has $\mathcal{J}(s) \to 1$ as $\Re s \to +\infty$, and hence $\Theta(s) \to 1/3$.

The associated Cayley transform is

$$\Theta(s) \;:=\; \frac{2\mathcal{J}(s) - 1}{2\mathcal{J}(s) + 1}. \tag{3}$$

Heuristically, $2\mathcal{J}$ plays the role of a Herglotz-type quantity and Θ the corresponding Schur function. The next section makes precise the key implication used later: a Schur bound on Θ forces removability of any isolated singularity of $\mathcal{J}$.

Lemma 4 (Zeros of ζ produce poles of $\mathcal{J}$)**.** *Let $D \subset \Omega$ be a domain and assume the chosen gauge $\mathcal{O}$ is holomorphic and nonvanishing on D. If $\rho \in D$ is a zero of $\zeta(s)$, then ρ is a pole of $\mathcal{J}(s)$ defined in* (2)*.*

Proof. By (2), the only possible singularities of $\mathcal{J}$ on D arise from zeros of ζ and from zeros of $\mathcal{O}$. The latter do not occur by assumption. The factor $(s-1)/s$ is holomorphic and nonzero on $D \subset \Omega$. Finally, $\det_2(I - A(s))$ is holomorphic and nonzero on Ω by (1). Hence a zero of ζ at ρ forces a pole of $\mathcal{J}$ at ρ. □

Schur and Herglotz classes (terminology). Let $D \subset \mathbb{C}$ be a domain. A holomorphic function Θ on D is called *Schur* if $|\Theta| \le 1$ on D. A holomorphic function H on D is called *Herglotz* if $\Re H \ge 0$ on D. The Cayley transform identifies these classes: if H is Herglotz and $H \not\equiv -1$, then

$$\Theta = \frac{H-1}{H+1}$$

is Schur. Conversely, if Θ is Schur and $\Theta \not\equiv 1$, then $(1+\Theta)/(1-\Theta)$ is Herglotz; see [4, 5].

Outline of the far-field strategy in this language. Theorem 1 will follow once we establish that Θ is Schur on $\{\,\Re s > 0.6\,\}$ (in the relevant gauge). Indeed, if $|\Theta| \le 1$ holds on $\{\,\Re s > 0.6\,\}$ away from the poles of $\mathcal{J}$, then Θ is bounded near any isolated singularity, and thus extends holomorphically across it by the removable singularity theorem. Since poles of $\mathcal{J}$ correspond to zeros of ζ in Ω by Lemma 4, this rules out zeros of ζ in the far region. The precise pinch argument is proved in the next section.

3. Schur/Herglotz pinch mechanism

This section records the analytic mechanism that converts a Schur bound for the Cayley field Θ into a zero-free region for ζ. The point is that a holomorphic function bounded by 1 cannot have a pole, and any isolated singularity is removable. In our setting, poles of $\mathcal{J}$ in Ω encode zeros of ζ (Lemma 4), so a Schur bound forces those zeros to be absent.

Removable singularities under a Schur bound.

Lemma 5 (Removable singularity under a strict Schur bound)**.** *Let $D \subset \mathbb{C}$ be a disc centered at ρ and let Θ be holomorphic on $D \setminus \{\rho\}$ with $|\Theta(s)| < 1$ there. Then Θ extends holomorphically to D and satisfies $|\Theta(s)| < 1$ for all $s \in D$. Consequently, the Cayley inverse*

$$\mathcal{H}(s) \;:=\; \frac{1+\Theta(s)}{1-\Theta(s)}$$

is holomorphic on D and satisfies $\Re\mathcal{H}(s) > 0$ on D.

Proof. Since Θ is bounded on the punctured disc, Riemann's removable singularity theorem gives a holomorphic extension to D (e.g. [7, Ch. 2]). The extension still satisfies $|\Theta| \le 1$ on D by continuity. If $|\Theta(\rho)| = 1$, then $|\Theta|$ attains its maximum at an interior point, so Θ is constant unimodular on D by the Maximum Modulus Principle (e.g. [7, Ch. 2]); this contradicts the strict bound $|\Theta| < 1$ on $D \setminus \{\rho\}$. Hence $|\Theta(\rho)| < 1$, and therefore $|\Theta| < 1$ holds everywhere on D. The Möbius map $w \mapsto (1+w)/(1-w)$ sends the unit disc into the right half-plane, so $\Re\mathcal{H} > 0$ on D. □

From a Schur bound to absence of poles.

Corollary 6 (Schur bound prevents poles of $\mathcal{J}$)**.** *Let $U \subset \Omega$ be a domain and let $S \subset U$ be a discrete set. Assume Θ is holomorphic on $U \setminus S$ and satisfies $|\Theta(s)| \le 1$ for all $s \in U \setminus S$. Then Θ extends holomorphically to U and satisfies $|\Theta| \le 1$ on U. If, moreover, Θ is not identically 1 on any connected component of U, then*

$$2\mathcal{J}(s) \;=\; \frac{1+\Theta(s)}{1-\Theta(s)}$$

extends holomorphically to U with $\Re(2\mathcal{J}) \ge 0$ there; in particular $\mathcal{J}$ has no poles in U.

Proof. Fix $\rho \in S$ and choose a disc $D \subset U$ centered at ρ with $D \cap S = \{\rho\}$. On $D \setminus \{\rho\}$ we have $|\Theta| \le 1$, hence Θ is bounded near ρ and extends holomorphically across ρ by Riemann's theorem (again [7, Ch. 2]). Doing this for each $\rho \in S$ yields a holomorphic extension of Θ to all of U. The bound $|\Theta| \le 1$ persists by continuity.

If $\Theta(s_0) = 1$ at an interior point s_0 of a connected component $U_0 \subset U$, then $|\Theta|$ attains its maximum at s_0, so Θ is constant unimodular on U_0 by the Maximum Modulus Principle [7, Ch. 2]. Thus the hypothesis excludes $\Theta \equiv 1$ on U_0, and therefore $1-\Theta$ is nonvanishing on U. Hence the Cayley inverse $(1+\Theta)/(1-\Theta)$ is holomorphic on U. Since the Cayley map sends the unit disc into the right half-plane, we have $\Re(2\mathcal{J}) \ge 0$ on U. A holomorphic function cannot have a pole, so $\mathcal{J}$ has no poles in U. □

Conclusion: Schur on a far half-plane implies Theorem 1. By Lemma 4, any zero ρ of ζ in Ω produces a pole of $\mathcal{J}$ in Ω (the numerator factors in (2) are nonzero on Ω).

On $\Omega = \{\Re s > \frac{1}{2}\}$ we have $0 \notin \Omega$, so the compensator $(s-1)/s$ introduces no pole on the working domain. The point $s = 1$ lies in Ω but the factor $(s-1)$ cancels the simple pole of ζ there, so $\mathcal{J}$ is holomorphic at $s = 1$ in the raw gauge $\mathcal{O} \equiv 1$. Whenever a nontrivial gauge $\mathcal{O}$ is introduced, we require (and, for bounds, separately verify) that $\mathcal{O}$ is holomorphic and nonvanishing on the stated domain; see Remark 3.

Therefore, if we can certify a Schur bound for Θ on a half-plane $U_\varepsilon = \{\,\Re s > 0.6 - \varepsilon\,\}$ with some $\varepsilon > 0$, then Corollary 6 implies $\mathcal{J}$ has no poles in U_ε, hence ζ has no zeros in U_ε. Since $\{\,\Re s \ge 0.6\,\} \subset U_\varepsilon$, this yields Theorem 1. The next section discharges the Schur bound by a boundary-certificate route and then specializes to U_ε.

4. All-heights Schur bound via a boundary wedge certificate

We now discharge the Schur bound required in Corollary 6 on a half-plane U_ε. The key input is an unconditional *boundary wedge* (P+) for a suitably outer-normalized version of $\mathcal{J}$ on the boundary line $\Re s = \frac{1}{2}$. This route is analytic (no large-height asymptotics) and applies for all heights.

Outer normalization on $\Re s = \frac{1}{2}$. Define

$$F(s) := \frac{\det_2(I - A(s))}{\zeta(s)} \cdot \frac{s-1}{s}, \qquad (\Re s > \tfrac{1}{2}),$$

and extend F to $\Omega \setminus Z(\zeta)$ by analytic continuation (removing the discrete pole set $Z(\zeta)$).

Lemma 7 (Boundary admissibility and Smirnov class for F)**.** *Let F be as above. Then on each connected component of $\Omega \setminus Z(\zeta)$:*

1. *F belongs to the Smirnov class N^+ (see, e.g., [9, Ch. 10]) and therefore admits nontangential boundary values $F^*(t) = \operatorname{nt-lim}_{\sigma \downarrow \frac{1}{2}} F(\sigma + it)$ for Lebesgue-a.e. $t \in \mathbb{R}$.*
2. *The boundary log-modulus $u(t) := \log|F^*(t)|$ lies in $L^1_{\mathrm{loc}}(\mathbb{R})$.*

Moreover, if $|u(t)| \le C\log(2+|t|)$ for $|t| \ge 1$, then $u \in L^1(\mathbb{R}, (1+t^2)^{-1}dt)$.

Proof. Fix a connected component U of $\Omega \setminus Z(\zeta)$. By Lemma 8, for every compact interval $I \Subset \mathbb{R}$ with $Q_\alpha(I) \Subset U$ the restriction of F to $Q_\alpha(I)$ is of bounded type. Since U is covered by such Whitney regions and bounded type is local on simply connected subdomains, it follows that F is of bounded type on U.

Next, on each such $Q_\alpha(I) \Subset U$, the boundary log-modulus of $\det_2(I-A)$ lies in $L^1(I)$ by Lemma 10, and $\log|\zeta(\frac{1}{2}+it)| \in L^1(I)$ with L^1-convergence from the interior by Lemma 11. Unwinding the definition of F (as a holomorphic combination of $\det_2(I-A)$ and ζ on U), this gives $\log|F^*| \in L^1_{\mathrm{loc}}$ on $\partial U \cap \{\Re s = \frac{1}{2}\}$. Applying Lemma 9 on each Whitney region yields $F \in N^+(U)$, hence F admits nontangential boundary values a.e. and $u(t) = \log|F^*(t)| \in L^1_{\mathrm{loc}}(\mathbb{R})$.

Finally, if $|u(t)| \le C\log(2+|t|)$ for $|t| \ge 1$, then

$$\int_{\mathbb{R}} \frac{|u(t)|}{1+t^2}\,dt \;\le\; C\int_{\mathbb{R}} \frac{\log(2+|t|)}{1+t^2}\,dt \;<\; \infty,$$

so $u \in L^1(\mathbb{R}, (1+t^2)^{-1}dt)$. □

Lemma 8 (Local bounded-type control for F from the Appendix normalizer)**.** *Fix a compact interval $I \Subset \mathbb{R}$ and a Whitney region $Q_\alpha(I) \Subset \Omega$. Assume the hypotheses of Appendix Lemma 28 and Theorem 29, so that the Appendix constructs an outer function $\mathcal{O}$ on Ω with boundary modulus $|\mathcal{O}(\frac{1}{2}+it)| = ig|\det_2(I-A(\frac{1}{2}+it))|\,ig|\xi(\frac{1}{2}+it)|^{-1}$ a.e. on I and defines $\mathcal{J}(s) = \det_2(I-A(s))/(\mathcal{O}(s)\,\xi(s))$ of bounded type on $Q_\alpha(I)$. Then F is of bounded type on $Q_\alpha(I)$.*

Proof. On $Q_\alpha(I)$, the Appendix constructs an outer function $\mathcal{O}$ with the stated boundary modulus on I and defines $\mathcal{J} = \det_2(I-A)/(\mathcal{O}\,\xi)$ of bounded type on $Q_\alpha(I)$. By the definition of F in the main text, F is obtained from $\mathcal{J}$ by composing with holomorphic operations that preserve bounded type on domains (products, quotients by nonvanishing bounded-type functions, and linear fractional transformations with holomorphic coefficients). Since $\mathcal{O}$ is outer and ξ is holomorphic and nonvanishing on $Q_\alpha(I) \subset \Omega \setminus Z(\zeta)$, these operations are legitimate on $Q_\alpha(I)$. Therefore F is of bounded type on $Q_\alpha(I)$. □

Lemma 9 (Smirnov upgrade from bounded type and boundary log-modulus)**.** *Let $U \subset \Omega$ be a simply connected domain with rectifiable boundary segment on $\Re s = \frac{1}{2}$ (e.g. a Whitney region $Q_\alpha(I)$ as in §A.1 of Appendix A). Let g be holomorphic on U and of bounded type (Nevanlinna class) on U. Assume g admits nontangential boundary values $g^*(t)$ for Lebesgue-a.e. t along $\partial U \cap \{\Re s = \frac{1}{2}\}$ and that $\log|g^*(t)| \in L^1_{\mathrm{loc}}(dt)$ on that boundary segment. Then $g \in N^+(U)$, and in particular g has nontangential boundary limits a.e. on $\partial U \cap \{\Re s = \frac{1}{2}\}$.*

Proof. By conformal mapping, it suffices to treat the case of the unit disk $\mathbb{D}$ (or upper half-plane) with boundary arc corresponding to the given rectifiable boundary segment. Since g is of bounded type on U, it belongs to the Nevanlinna class on U; equivalently, $g = h/k$ with $h, k \in H^\infty(U)$ and $k \not\equiv 0$. The hypothesis $\log|g^*| \in L^1_{\mathrm{loc}}$ on the boundary segment implies that the boundary values of $\log|k^*|$ are locally integrable there as well (because h is bounded), so the outer-function construction on U produces an outer function k_{out} with $|k^*_{\mathrm{out}}| = |k^*|$ a.e. on that segment. Replacing k by k_{out} and h by $h\,k/k_{\mathrm{out}}$ (which remains bounded and holomorphic) yields a representation $g = \tilde{h}/k_{\mathrm{out}}$ with $\tilde{h} \in H^\infty(U)$ and k_{out} outer. This is precisely $g \in N^+(U)$. In particular, functions in $N^+(U)$ admit nontangential boundary limits a.e. on the corresponding boundary segment. □

Lemma 10 (From Carleson energy to L^1 boundary control for $\log|\det_2|$)**.** *Fix a compact interval $I \Subset \mathbb{R}$ and $\varepsilon_0 \in (0, \frac{1}{2}]$. Let*

$$U_{\det_2}(\sigma, t) = \log\Big|\det_2\Big(I - A(\tfrac{1}{2}+\sigma+it)\Big)\Big|, \qquad (\sigma,t) \in (0,\varepsilon_0] \times I,$$

where $\log|\det_2(I-A)|$ is interpreted componentwise as the real part of any analytic branch $\mathrm{Log}(\det_2(I-A))$ on each connected component of $\Omega \setminus Z(\det_2(I-A))$ (so it is branch-independent).

Further, $\log|\det_2(I-A)|$ *is subharmonic on* Ω *and harmonic on* $\Omega\setminus Z(\det_2(I-A))$*; since the (discrete) zero set is polar, it does not affect harmonic-measure boundary trace statements used below. Assume the Carleson energy bound of Lemma 24 for* $\nabla U_{\det_2}$ *on* $Q(I)$*, uniformly up to height* ε_0*. Then the boundary trace* $u_{\det_2}(t) := \lim_{\sigma\downarrow 0} U_{\det_2}(\sigma,t)$ *exists in* $\mathrm{BMO}(I)$ *(hence in* $L^1(I)$*), and in particular*

$$\sup_{0<\sigma\le\varepsilon_0} \|U_{\det_2}(\sigma,\cdot)\|_{L^1(I)} \;<\; \infty.$$

Proof. On $\Omega\setminus Z(\det_2(I-A))$, the function $U_{\det_2} = \log|\det_2(I-A)|$ is harmonic. The Carleson energy hypothesis implies that the measure $|\nabla U_{\det_2}(\sigma,t)|^2\,\sigma\,d\sigma\,dt$ is Carleson on $Q(I)$. By the Fefferman–Stein characterization of BMO boundary traces via Carleson measures for Poisson/harmonic extensions (see, for example, [10, Ch. IV, Thm. 3, p. 159] and [11, Ch. VI, Thm. 3.4]), $U_{\det_2}$ admits nontangential boundary values $U^*_{\det_2} \in \mathrm{BMO}(I)$, hence $U^*_{\det_2} \in L^1(I)$. In particular, $U_{\det_2}(\sigma,\cdot) \to U^*_{\det_2}$ in $L^1(I)$ as $\sigma\downarrow 0$. Moreover, $U_{\det_2}(\sigma,\cdot)$ admits a nontangential boundary trace in $\mathrm{BMO}(I)$ (hence in $L^1(I)$); see [10, 11]. and depends only on the modulus, hence is independent of any choice of analytic branch for $\mathrm{Log}(\det_2(I-A))$. The Carleson energy hypothesis in Lemma 24 gives a Carleson-measure bound for $|\nabla U_{\det_2}|^2\,\sigma\,d\sigma\,dt$ over the Carleson box above I. By the Carleson-measure characterization of BMO boundary traces for harmonic functions on the upper half-plane, this implies that the nontangential boundary trace $u_{\det_2}(t) = \lim_{\sigma\downarrow 0} U_{\det_2}(\sigma,t)$ exists in $\mathrm{BMO}(I)$; in particular $u_{\det_2} \in L^1(I)$. Moreover, the same characterization yields the uniform $L^1(I)$ control $\sup_{0<\sigma\le\varepsilon_0}\|U_{\det_2}(\sigma,\cdot)\|_{L^1(I)} < \infty$. Since the zero set $Z(\det_2(I-A))$ is discrete (hence polar), removing it does not affect harmonic-measure boundary trace statements. □

Lemma 11 (Boundary log-modulus control for ζ on components)**.** *Fix a compact interval* $I \Subset \mathbb{R}$ *and* $\varepsilon_0 \in (0,\frac{1}{2}]$*. Let* U *be a connected component of* $\Omega\setminus Z(\zeta)$ *intersecting* $Q_{\varepsilon_0}(I)$*. Then* ζ *is holomorphic and nonvanishing on* U*, hence* $u(s) = \log|\zeta(s)|$ *is harmonic on* U*. Moreover, the boundary trace* $t \mapsto \log|\zeta(\frac{1}{2}+it)|$ *lies in* $L^1(I)$ *and*

$$\log|\zeta(\tfrac{1}{2}+\varepsilon+it)| \to \log|\zeta(\tfrac{1}{2}+it)| \quad \text{in } L^1(I) \text{ as } \varepsilon\downarrow 0.$$

Proof. Let U be a connected component of $\Omega\setminus Z(\zeta)$ intersecting $Q_{\varepsilon_0}(I)$. Then ζ is holomorphic and nonvanishing on U, hence $u(s) = \log|\zeta(s)|$ is harmonic on U. On the compact strip segment $\{\sigma+it : \sigma\in[\frac{1}{2},\frac{1}{2}+\varepsilon_0],\ t\in I\}$, ζ has only finitely many zeros (counted with multiplicity). For each zero s_k in this compact set, write $\zeta(s) = (s-s_k)^{m_k} g_k(s)$ with g_k holomorphic and nonvanishing in a neighborhood of s_k. Covering the compact strip by finitely many such neighborhoods and a zero-free remainder shows that on the strip

$$\log|\zeta(s)| = \sum_k m_k \log|s-s_k| + O(1),$$

with the $O(1)$ bounded on the strip. For each fixed s_k, the functions $t\mapsto \log|(\frac{1}{2}+\varepsilon+it)-s_k|$ are uniformly $L^1(I)$-bounded for $\varepsilon\in(0,\varepsilon_0]$ and converge in $L^1(I)$ as $\varepsilon\downarrow 0$. Therefore dominated convergence yields the stated $L^1(I)$ convergence $\log|\zeta(\frac{1}{2}+\varepsilon+it)| \to \log|\zeta(\frac{1}{2}+it)|$ as $\varepsilon\downarrow 0$. □

Lemma 12 (Local L^1 control of $\log|F^*|$ on boundary intervals)**.** *Fix a compact interval* $I \Subset \mathbb{R}$ *and* $\varepsilon_0 \in (0,\frac{1}{2}]$*, and set*

$$Q_{\varepsilon_0}(I) := \{\, \tfrac{1}{2}+\sigma+it :\ 0<\sigma\le\varepsilon_0,\ t\in I \,\} \Subset \Omega.$$

Let

$$F(s) := \det_2(I-A(s))\,\frac{s-1}{s\,\zeta(s)}, \qquad s\in\Omega\setminus Z(\zeta).$$

Assume:

(i) $\log|\det_2(I-A(\frac{1}{2}+\varepsilon+it))| \in L^1(I)$ *uniformly for* $\varepsilon\in(0,\varepsilon_0]$*, and the nontangential boundary limit* $\log|\det_2(I-A(\frac{1}{2}+it))|$ *exists in* $L^1(I)$*;*

(ii) *for each connected component U of $\Omega\setminus Z(\zeta)$ intersecting $Q_{\varepsilon_0}(I)$, the function $\log|\zeta(\frac{1}{2}+\varepsilon+it)|$ has an $L^1(I)$-limit as $\varepsilon\downarrow 0$ when restricted to U.*

Then on each such component U, the nontangential boundary values $F^*(t)$ exist for Lebesgue-a.e. $t\in I$, and $\log|F^*(t)|\in L^1_{\mathrm{loc}}(I)$ on U.

Proof. Fix a component U as in the statement. For $s=\frac{1}{2}+\varepsilon+it$ with $0<\varepsilon\le\varepsilon_0$ and $t\in I$, we have

$$\log|F(s)| = \log|\det_2(I-A(s))| + \log|s-1| - \log|s| - \log|\zeta(s)|.$$

Since I is compact and $\varepsilon\in(0,\varepsilon_0]$, the functions $t\mapsto\log|\frac{1}{2}+\varepsilon+it|$ and $t\mapsto\log|-\frac{1}{2}+\varepsilon+it|$ are bounded on I, uniformly in ε; hence $\log|s|$ and $\log|s-1|$ contribute uniformly bounded $L^1(I)$ terms. Assumptions (i)–(ii) therefore imply that $\log|F(\frac{1}{2}+\varepsilon+it)|$ is uniformly in $L^1(I)$ and has an $L^1(I)$ limit as $\varepsilon\downarrow 0$ along U. In particular, after passing to a subsequence if needed, $F(\frac{1}{2}+\varepsilon+it)$ has a nontangential boundary limit for a.e. $t\in I$, and the limiting boundary modulus satisfies $\log|F^*(t)|\in L^1_{\mathrm{loc}}(I)$ on U. □

Lemma 13 (Outer factor from boundary modulus on Ω)**.** *Assume Lemma 7 together with $u\in L^1(\mathbb{R},(1+t^2)^{-1}dt)$. Then there exists a holomorphic function $\mathcal{O}_\zeta$ on Ω, unique up to a unimodular constant, with no zeros on Ω, such that the nontangential boundary values satisfy*

$$ig|\mathcal{O}_\zeta(\tfrac{1}{2}+it)| = ig|F^*(t)| \qquad \text{for Lebesgue-a.e. } t\in\mathbb{R}.$$

Moreover, $\log|\mathcal{O}_\zeta(s)|$ is the Poisson extension of $u(t)$ from the boundary line $\Re s=\frac{1}{2}$.

Proof. Translate Ω to the right half-plane $\{\,\Re w>0\,\}$ via $w=s-\frac{1}{2}$. Since $u\in L^1(\mathbb{R},(1+t^2)^{-1}dt)$, its Poisson extension $U=\mathcal{P}[u]$ is a harmonic function on Ω with nontangential boundary trace u a.e. Choose a harmonic conjugate V of U on Ω and set $\mathcal{O}_\zeta:=\exp(U+iV)$. Then $\mathcal{O}_\zeta$ is holomorphic and zero-free on Ω, and by Fatou theory its boundary modulus is $e^{u(t)}$ for a.e. t. Uniqueness up to a unimodular constant follows because the ratio of two such outer functions has boundary modulus 1 a.e. and hence is an inner constant; see Garnett [11, Ch. II]. □

Define the outer-normalized ratio

$$\mathcal{J}_{\mathrm{out}}(s) := \frac{F(s)}{\mathcal{O}_\zeta(s)} = \frac{\det_2(I-A(s))}{\mathcal{O}_\zeta(s)\,\zeta(s)}\cdot\frac{s-1}{s}. \tag{4}$$

Then $|\mathcal{J}_{\mathrm{out}}(\frac{1}{2}+it)|=1$ for Lebesgue-a.e. t.

Boundary wedge (P+). Let $w(t):=\operatorname{Arg}\mathcal{J}_{\mathrm{out}}(\frac{1}{2}+it)$ be the boundary phase (defined for a.e. t). We say that (P+) holds if there exists $m\in\mathbb{R}$ such that

$$|w(t)-m|<\frac{\pi}{2} \qquad \text{for Lebesgue-a.e. } t\in\mathbb{R}.$$

Equivalently, $\Re\!\mathrm{e}\,g(e^{-im}\mathcal{J}_{\mathrm{out}}(\frac{1}{2}+it))\ge 0$ for Lebesgue-a.e. t.

Theorem 14 (Boundary wedge (P+))**.** *The boundary wedge* (P+) *holds for $\mathcal{J}_{\mathrm{out}}$.*

Proof. See §§A.3–A.5 in Appendix A. □

From (P+) to an interior Schur bound. Fix m witnessing (P+) and set

$$\widetilde{\mathcal{J}}(s) := e^{-im}\mathcal{J}_{\mathrm{out}}(s).$$

Then $|\widetilde{\mathcal{J}}(\frac{1}{2}+it)|=1$ a.e. and $\Re\,\widetilde{\mathcal{J}}(\frac{1}{2}+it)\ge 0$ a.e. Define the corresponding Cayley field

$$\Theta_{\mathrm{out}}(s) := \frac{2\widetilde{\mathcal{J}}(s)-1}{2\widetilde{\mathcal{J}}(s)+1}.$$

Lemma 15 (Smirnov regularity for $\mathcal{J}_{\text{out}}$ and Θ_{out})**.** *Assume Lemmas 7 and 13. Then $\mathcal{J}_{\text{out}} \in N^+(\Omega\setminus Z(\zeta))$ and admits nontangential boundary values $\mathcal{J}_{\text{out}}(\frac{1}{2}+it)$ for Lebesgue-a.e. t. Consequently, $\widetilde{\mathcal{J}} = e^{-im}\mathcal{J}_{\text{out}} \in N^+(\Omega \setminus Z(\zeta))$, and Θ_{out} admits Lebesgue-a.e. boundary values on $\Re s = \frac{1}{2}$.*

Proof. By Lemma 7, $F \in N^+(\Omega \setminus Z(\zeta))$. By Lemma 13, $\mathcal{O}_\zeta$ is holomorphic, zero-free, and outer on Ω, hence lies in $N^+(\Omega)$. Therefore $\mathcal{J}_{\text{out}} = F/\mathcal{O}_\zeta \in N^+(\Omega \setminus Z(\zeta))$. Multiplication by a unimodular constant preserves N^+ membership, so $\widetilde{\mathcal{J}} \in N^+(\Omega \setminus Z(\zeta))$. The boundary-value statements follow from Smirnov boundary theory (e.g. Garnett [11, Ch. II]). Finally, Θ_{out} is a Möbius transform of $\widetilde{\mathcal{J}}$; its a.e. boundary values exist wherever $2\widetilde{\mathcal{J}}(\frac{1}{2}+it) \neq -1$, which holds a.e. since $\Re\,\widetilde{\mathcal{J}}(\frac{1}{2}+it) \geq 0$ a.e. □

Lemma 16 (Boundary-to-interior Schur transport on components of $\Omega\setminus Z(\zeta)$)**.** *Let U be a connected component of $\Omega \setminus Z(\zeta)$ and let $\Theta \in N^+(U)$ admit nontangential boundary values $\Theta(\frac{1}{2}+it)$ for Lebesgue-a.e. t. If $|\Theta(\frac{1}{2}+it)| \leq 1$ for Lebesgue-a.e. t, then $|\Theta(s)| \leq 1$ for all $s \in U$.*

Proof. For $\Theta \not\equiv 0$, the function $u := \log|\Theta|$ is subharmonic on U and has a harmonic majorant because $\Theta \in N^+(U)$. At Lebesgue points where the nontangential boundary values exist, the boundary hypothesis gives $u(\frac{1}{2}+it) \leq 0$ a.e. Since $Z(\zeta)$ is discrete, it is a polar set, hence has harmonic measure zero for each component U; therefore the Poisson/harmonic-measure domination principle on U yields $u(s) \leq 0$ for all $s \in U$. Thus $|\Theta(s)| \leq 1$ on U. See Garnett [11, Ch. II] and Ransford [8, Ch. 5]. □

Lemma 17 (Boundary-to-interior Herglotz transport on components of $\Omega \setminus Z(\zeta)$)**.** *Let U be a connected component of $\Omega \setminus Z(\zeta)$ and let $G \in N^+(U)$ admit nontangential boundary values $G(\frac{1}{2}+it)$ for Lebesgue-a.e. t. If $\Re\,G(\frac{1}{2}+it) \geq 0$ for Lebesgue-a.e. t, then $\Re\,G(s) \geq 0$ for all $s \in U$.*

Proof. Apply Lemma 16 to the Cayley transform $\Phi(s) := \frac{G(s)-1}{G(s)+1}$ (defined wherever $G \neq -1$) and use continuity to conclude $\Re\,G \geq 0$ on U. Equivalently, one may apply the harmonic-measure domination principle directly to the harmonic function $\Re\,G$ using that $G \in N^+(U)$. □

Proposition 18 (Herglotz/Schur transport)**.** *Assume* (P+) *for $\mathcal{J}_{\text{out}}$ and Lemma 15. Then Θ_{out} is Schur on each connected component of $\Omega \setminus Z(\zeta)$, and*

$$H(s) := \frac{1+\Theta_{\text{out}}(s)}{1-\Theta_{\text{out}}(s)}$$

is Herglotz on $\Omega \setminus Z(\zeta)$ (i.e. $\Re H(s) \geq 0$ there). Moreover, $H(s) = 2\widetilde{\mathcal{J}}(s) = 2e^{-im}\mathcal{J}_{\text{out}}(s)$.

Proof. Let U be any connected component of $\Omega \setminus Z(\zeta)$. By Lemma 15 we have $\widetilde{\mathcal{J}} \in N^+(U)$ and it admits nontangential boundary values on $\Re s = \frac{1}{2}$ for Lebesgue-a.e. t. By (P+) (with the fixed phase m), $\Re\,\widetilde{\mathcal{J}}(\frac{1}{2}+it) \geq 0$ for Lebesgue-a.e. t. Therefore Lemma 17 yields $\Re\,\widetilde{\mathcal{J}}(s) \geq 0$ for all $s \in U$. Since the Cayley map $z \mapsto \frac{2z-1}{2z+1}$ sends the closed right half-plane into the closed unit disc, it follows that $|\Theta_{\text{out}}(s)| \leq 1$ for all $s \in U$. As U was arbitrary, Θ_{out} is Schur on each connected component of $\Omega \setminus Z(\zeta)$. Finally, the Cayley inverse gives

$$H(s) = \frac{1+\Theta_{\text{out}}(s)}{1-\Theta_{\text{out}}(s)} = 2\widetilde{\mathcal{J}}(s) = 2e^{-im}\mathcal{J}_{\text{out}}(s),$$

so H is Herglotz on $\Omega \setminus Z(\zeta)$. □

4.1. **Proof of the main theorem.**

Proof of Theorem 1 (direct contradiction). Suppose for contradiction that $\zeta(\rho_0) = 0$ with $\rho_0 = \beta_0 + i\gamma_0$ and $\beta_0 \geq 0.6$. Set $\delta_0 := \beta_0 - \frac{1}{2} \geq 0.1$.

Choice of Whitney parameter. Let $A := Z_0\, C_{\text{test}}\, \sqrt{2C_{\text{box}}(\alpha')}$ be the structural constant from the CR–Green and energy bounds (depending only on α' and the window ψ; see (6)). Choose a fixed

$$c \;\in\; \Big(0,\; \frac{(11/A)^2\, \log\sqrt{2}}{1}\Big), \qquad L \;:=\; \min\Big\{\frac{c}{\log\langle\gamma_0\rangle},\; 1\Big\}.$$

Then $L \le 1$ always, and $A\sqrt{L} \le A\sqrt{c/\log\sqrt{2}} < 11$.

Step 1 (phase-velocity lower bound). By Theorem 29, the distributional boundary-phase derivative $-w'$ of $\mathcal{J}_{\text{out}}$ is a positive measure that includes the Poisson balayage of every off-critical zero. In particular, $-w'$ dominates the term $2\delta_0/(\delta_0^2 + (t-\gamma_0)^2)$ coming from ρ_0. Center a window ψ_{L,γ_0} at γ_0. Since $\psi_{L,\gamma_0} \equiv 1$ on $[\gamma_0 - L, \gamma_0 + L]$, $L \le 1$, and $\delta_0 \ge 0.1$:

$$\int_{\mathbb{R}} \psi_{L,\gamma_0}(t)\,(-w'(t))\,dt \;\ge\; 4\pi \arctan(L/\delta_0) \;\ge\; \frac{4\pi L}{\delta_0 + L} \;\ge\; \frac{4\pi L}{1.1} \;>\; 11\,L. \tag{5}$$

Step 2 (CR–Green upper bound). By Proposition 38 applied to the admissible test $\varphi_{L,\gamma_0} = Z_0^{-1} L^{-1} \psi_{L,\gamma_0}$ and then multiplied by $Z_0 L$:

$$\int_{\mathbb{R}} \psi_{L,\gamma_0}\,(-w') \;\le\; Z_0\, C_{\text{test}}\, \sqrt{E(I)} \cdot L.$$

Proposition 33 gives the **height-independent** bound $E(I) \le C_{\text{box}}\,|I|$ (using the Carleson measure theorem for the inner reciprocal $\mathcal{I}$ from Lemma 31). Hence $\sqrt{E(I)} \le \sqrt{C_{\text{box}}|I|} = \sqrt{2C_{\text{box}}L}$, and

$$Z_0\, C_{\text{test}}\, \sqrt{E(I)} \cdot L \;\le\; Z_0\, C_{\text{test}}\, \sqrt{2C_{\text{box}}}\; L^{3/2} \;=:\; A\, L^{3/2}, \tag{6}$$

where $A := Z_0\, C_{\text{test}}\, \sqrt{2C_{\text{box}}(\alpha', c)}$ depends only on (α', c, ψ).

Step 3 (contradiction). Combining (5) and (6): $11\,L \le A\,L^{3/2}$, hence $11 \le A\,L^{1/2}$. Since $L = c/\log\langle\gamma_0\rangle \le c/\log\sqrt{2}$, choosing c so that $A\sqrt{c/\log\sqrt{2}} < 11$ (i.e. $c < (11/A)^2 \log\sqrt{2}$) forces a contradiction. Therefore no such ρ_0 exists, and $\zeta(s) \ne 0$ for $\Re s \ge 0.6$.

Small-height case ($|\gamma_0| \le T_0$). For any finite height T_0, the finite region $\{|\gamma| \le T_0,\ \Re s \ge 0.6\}$ is covered by the shipped ball-arithmetic rectangle certificate (Table 1) for $T_0 = 20$. Alternatively, for $|\gamma_0| \le T_0$ one may choose c small enough that $L \le 1$ and $A\sqrt{L} < 11$ hold simultaneously. □

Remark 19 (Alternative route via the boundary wedge (P+)). The same conclusion follows (with the same inputs) by the boundary-wedge route of §4–Appendix A: Theorem 14 ⇒ Proposition 18 ⇒ Corollary 6. The direct-contradiction proof above is logically simpler because it avoids the global boundary-wedge statement (P+), the Smirnov boundary transport, and the interior Herglotz/Schur mechanism; it needs only the *localized* phase-velocity lower bound from a single hypothetical zero.

TABLE 1. Supplementary computational artifacts (not used in the proof).

Artifact	**Parameter**	**Value**
Rectangle certification (`theta_certify`)		
Domain	$[\sigma_{\min}, \sigma_{\max}] \times [t_{\min}, t_{\max}]$	$[0.6, 0.7] \times [0, 20]$
upper bound	$\max \lvert\Theta_{\text{proj}}\rvert$	0.9999928763
Safety margin	$1 - \theta_{\text{hi}}$	7.12×10^{-6}
Status	`ok`	`true`
Boxes processed		380,764
Precision	(bits)	260
Gauge		`outer_zeta_proj`
Pick certificate (`pick_certify`, $\sigma_0 = 0.599$)		
Matrix size	N	16
Spectral separation	δ_{cert}	0.594
SPD at origin	$P_N \succ 0$	`true`
Coefficient count	N_{coeff}	128
Tail sum (diagnostic)	$\sum_{16}^{127} \lvert a_n\rvert$	0.67
Gauge		`raw_zeta`
Pick certificate (`pick_certify`, $\sigma_0 = 0.6$)		
Matrix size	N	16
Spectral separation	δ_{cert}	0.594
SPD at origin	$P_N \succ 0$	`true`
Coefficient count	N_{coeff}	128
Tail sum (diagnostic)	$\sum_{16}^{127} \lvert a_n\rvert$	0.67
Gauge		`raw_zeta`
Pick certificate (`pick_certify`, $\sigma_0 = 0.7$)		
Matrix size	N	16
Spectral separation	δ_{cert}	0.627
SPD at origin	$P_N \succ 0$	`true`
Coefficient count	N_{coeff}	128
Tail sum (diagnostic)	$\sum_{16}^{127} \lvert a_n\rvert$	0.61
Gauge		`raw_zeta`

ottomrule

CONCLUSION AND LIMITATIONS (UNCONDITIONAL STATUS)

We prove a fixed half-plane zero-exclusion for the Riemann zeta function: $\zeta(s) \neq 0$ for $\Re s \geq 0.6$ (Theorem 1).

The argument is analytic and self-contained. Zeros of ζ in Ω are converted into zeros of the analytic inner reciprocal $\mathcal{I} := B^2/\mathcal{J}_{\text{out}}$ (Lemma 31), whose non-circular boundedness $|\mathcal{I}| \leq 1$ follows from the Phragmén–Lindelöf principle (no assumption about ζ-zeros is used). The resulting nonnegative potential $W = -\log|\mathcal{I}| \geq 0$ provides a Whitney-box energy bound (Proposition 33) with growth $E(I) \leq C \log^2\langle t_0\rangle\, |I|$. The direct-contradiction proof of Theorem 1 chooses the Whitney parameter $c = c_0/\log\langle\gamma_0\rangle$ (proportional to $1/\log$ of the height of the hypothetical zero), causing the $\log^2$ factor in the energy to cancel against $|I| \propto 1/\log^2$, yielding a height-independent ratio of upper to lower bounds that gives a contradiction for a universal choice of $c_0 < (11/A)^2/2$.

Computer assistance and verifiability. The core implications are analytic. The repository provides supplementary rigorous numerical certificates (ball arithmetic) in Table 1 for low-height cross-checks; these are included for transparency but are not logically required for the all-heights proof (except possibly for the small-height case $|\gamma_0| < 2$, where the explicit constants have not been tracked; see the proof of Theorem 1).

Limitations and scope. We do not claim the Riemann Hypothesis here.
Computer assistance and verifiability. Although the core implications are analytic, the repository provides rigorous numerical certificates (ball arithmetic) together with a verifier and JSON outputs so that the finite checks can be independently verified.
Limitations and scope. We do not claim the Riemann Hypothesis here.

All conclusions are confined to the explicit region proved, and no statement is made about zeros with $\Re s < 0.6$. It isolates and certifies a fixed far-field exclusion $\Re s \geq 0.6$. Pushing the boundary 0.6 closer to 1/2 within this framework would require sharpening the analytic boundary-certificate constants and the Carleson/box-energy bounds that enter the wedge criterion, which we do not pursue here.

Statements and Declarations

Competing interests. The author declares no competing interests.
Data and materials availability. For reproducibility, the verification is with respect to the shipped bundle (and its SHA-256 manifest, if provided). If any file is updated after theory revision, the verification should be rerun and the manifest regenerated. All computational artifacts used for supplementary cross-checks are included in the handoff bundle (and mirrored in the repository):

```
compute/artifacts/theta_certify_sigma06_07_t0_20_outer_zeta_proj.json
compute/artifacts/pick_sigma0599_raw_zeta_N16.json
compute/artifacts/pick_sigma06_raw_zeta_N16.json
compute/artifacts/pick_sigma07_raw_zeta_N16.json
compute/verify_attachment_arb.py
```

Remark 20. The verifier is based on rigorous ball arithmetic (ARB via `python-flint`) and is intended to be independently verifiable. Appendix B records the verification manifest and a minimal command sequence for running the verifier on a fresh machine.

Appendix A. Proof of the boundary wedge certificate (P+)

This appendix proves Theorem 14. The argument combines a quantitative phase–velocity identity with a Whitney-box Cauchy–Riemann/Green pairing and a Carleson-type energy bound, yielding a windowed phase estimate that implies the boundary wedge inclusion (P+).

A.1. **Statement, standing notation, and domains.** This subsection fixes the ambient domain, boundary conventions, Whitney geometry, and the meaning of boundary limits, so later phase and energy identities are unambiguous.

Throughout Appendix A we work in the right half-plane

$$\Omega := \{s \in \mathbb{C} : \Re s > \tfrac{1}{2}\},$$

with boundary line $\partial\Omega = \{\frac{1}{2} + it : t \in \mathbb{R}\}$. All analytic objects are understood componentwise on $\Omega \setminus Z$, where Z denotes the relevant zero/pole set, so that branches of log and Arg are well-defined on each connected component.

For a compact interval $I \subset \mathbb{R}$ and a dilation parameter $\alpha > 1$ we write $Q_\alpha(I)$ for the Whitney box based on I, and we use the weighted area measure $\sigma\, dt\, d\sigma$ on Ω, where $\sigma := \Re s - \frac{1}{2}$.

The goal is to prove the boundary wedge certificate (P+) stated in Theorem 14. The proof proceeds by: (i) a quantitative phase–velocity identity for the boundary phase of $\mathcal{J}_{\text{out}}$, (ii) a Green/Cauchy–Riemann pairing on Whitney boxes, (iii) a Carleson-type energy bound for a logarithmic derivative, and (iv) a quantitative wedge criterion converting windowed phase control into a.e. wedge inclusion.

In the phase–velocity identity below, we will show that $-w'$ is positive (in the sense of distributions): a locally finite measure plus a discrete atomic part encoding the off–critical zero data that enter the boundary phase derivative.

A.2. **A quantitative wedge criterion from Whitney-local control.** We state the wedge target (P+) in a form suited to local Whitney-box estimates and record the boundary conventions used throughout Appendix A.

We work on the boundary line $\Re s = \frac{1}{2}$ and use the following conventions.

- *Wedge.* For an aperture parameter $\alpha \in (0, \frac{\pi}{2})$ and a center angle $m \in \mathbb{R}$, write
$$W_{m,\alpha} := \{z \in \mathbb{C} : \ |\operatorname{Arg}(e^{-im}z)| \le \alpha\}.$$
Thus (P+) is the Lebesgue-a.e. inclusion $\mathcal{J}_{\text{out}}(\frac{1}{2}+it) \in W_{m,\alpha}$ for some fixed $\alpha < \frac{\pi}{2}$ and some $m \in \mathbb{R}$.
- *Whitney / Carleson boxes.* For an interval $I \subset \mathbb{R}$, write the Carleson box $S(I) := \{\frac{1}{2}+\sigma+it : 0 < \sigma \le |I|, \ t \in I\}$. A Whitney box means a box of comparable width and height, e.g. $\{\frac{1}{2}+\sigma+it : \ \sigma \in [a|I|, b|I|], \ t \in I\}$ with fixed $0 < a < b$.
- *Meaning of "a.e."* Unless explicitly stated otherwise, "a.e." refers to Lebesgue measure dt on $\mathbb{R}$.

Lemma 21 (Outer normalizer from boundary log-modulus)**.** *Let $u \in L^1(\mathbb{R}, (1+t^2)^{-1}dt)$ be real-valued. Then there exists an outer function O on Ω (zero-free and holomorphic on Ω) whose nontangential boundary values satisfy*
$$|O(\tfrac{1}{2}+it)| = e^{u(t)} \quad \textit{for a.e. } t \in \mathbb{R}.$$
Moreover O is unique up to a unimodular constant.

Proof. Define the Poisson extension U of u to Ω by
$$U(\tfrac{1}{2}+\sigma+it) \ := \ \frac{1}{\pi}\int_{\mathbb{R}} u(\tau)\,\frac{\sigma}{\sigma^2+(t-\tau)^2}\,d\tau, \qquad \sigma > 0.$$
The weighted integrability $u \in L^1(\mathbb{R}, (1+t^2)^{-1}dt)$ ensures the integral converges and that U is harmonic on Ω. Let V be a harmonic conjugate of U on Ω (defined up to an additive constant), and set
$$O(s) \ := \ \exp ig(U(s)+iV(s)).$$
Then O is holomorphic and zero-free on Ω. By the nontangential boundary limit theorem for Poisson extensions of L^1_{loc} boundary data, one has $U(\frac{1}{2}+\varepsilon+it) \to u(t)$ for a.e. t as $\varepsilon \downarrow 0$; hence the nontangential boundary values satisfy $|O(\frac{1}{2}+it)| = e^{u(t)}$ for a.e. t; see Duren [9, Ch. II] or Garnett [11, Ch. II]. Uniqueness up to unimodular constant follows because the ratio of two such outer functions has a.e. boundary modulus 1 and hence is an inner constant. (Source.) This is the outer-function construction in half-planes; see Duren *H^p Spaces*, Ch. II, or Garnett *Bounded Analytic Functions*, Ch. II. □

A.3. **Phase–velocity identity (quantitative form) and boundary passage.** We establish the boundary phase–velocity relation for the outer-normalized ratio, and record the precise sense in which derivatives and boundary traces are taken.

Lemma 22 (Outer–Hilbert boundary identity)**.** *Let $u \in L^1_{\text{loc}}(\mathbb{R})$ and let O be an outer function on Ω whose boundary modulus satisfies $|O(\frac{1}{2}+it)| = e^{u(t)}$ for a.e. t. Let $w(t) := \operatorname{Arg} O(\frac{1}{2}+it)$ denote the boundary argument (defined modulo an additive constant). Then, in $\mathcal{D}'(\mathbb{R})$,*
$$\frac{d}{dt}w(t) = \mathcal{H}[u'](t),$$
where $\mathcal{H}$ is the boundary Hilbert transform on $\mathbb{R}$ (as a continuous operator $\mathcal{D}'(\mathbb{R}) \to \mathcal{D}'(\mathbb{R})$) and u' is the distributional derivative.

Proof. Write $\log O = U + iV$ on Ω, where $U = \Re \log O$ is harmonic and $V = \Im \log O$ is its harmonic conjugate (on each component of $\Omega \setminus Z(O)$, fixing a branch of log). The boundary trace satisfies $U(\frac{1}{2} + \cdot) = u$ in $\mathcal{D}'(\mathbb{R})$, and the conjugate boundary trace is $V(\frac{1}{2} + \cdot) = \mathcal{H}[u]$ in $\mathcal{D}'(\mathbb{R})$ (up to an additive constant). Differentiating in t in the sense of distributions gives

$$\frac{d}{dt} \operatorname{Arg} O(\tfrac{1}{2} + it) = \partial_t V(\tfrac{1}{2} + it) = \mathcal{H}[\partial_t u](t) = \mathcal{H}[u'](t),$$

since differentiation commutes with $\mathcal{H}$ on $\mathcal{D}'(\mathbb{R})$. □

Lemma 23 (Smoothed distributional bound for $\partial_\sigma \Re \log \det_2$)**.** *Let $I \Subset \mathbb{R}$ be a compact interval and fix $\varepsilon_0 \in (0, \frac{1}{2}]$. There exists a finite constant*

$$C_* \;:=\; \sum_p \sum_{k \geq 2} \frac{p^{-k/2}}{k^2 \, \log p} \;<\; \infty$$

such that for all $\sigma \in (\frac{1}{2}, \frac{1}{2} + \varepsilon_0]$ and every $\varphi \in C_c^2(I)$,

$$\Big| \int_{\mathbb{R}} \varphi(t) \, \partial_\sigma \Re \log \det_2 \big(I - A(\sigma + it)\big) \, dt \Big| \;\leq\; C_* \, \|\varphi''\|_{L^1(I)}.$$

Proof. For $\sigma > \frac{1}{2}$ one has the absolutely convergent expansion

$$\partial_\sigma \, \Re \log \det_2 \big(I - A(\sigma + it)\big) \;=\; \sum_p \sum_{k \geq 2} (\log p) \, p^{-k\sigma} \cos(kt \log p).$$

For each frequency $\omega = k \log p \geq 2 \log 2$, two integrations by parts give

$$\Big| \int_{\mathbb{R}} \varphi(t) \cos(\omega t) \, dt \Big| \;\leq\; \frac{\|\varphi''\|_{L^1(I)}}{\omega^2}.$$

Summing the resulting majorant yields

$$\Big| \int \varphi \, \partial_\sigma \Re \log \det_2 \, dt \Big| \;\leq\; \|\varphi''\|_{L^1} \sum_p \sum_{k \geq 2} \frac{(\log p) \, p^{-k\sigma}}{(k \log p)^2} \;\leq\; \|\varphi''\|_{L^1} \sum_p \sum_{k \geq 2} \frac{p^{-k/2}}{k^2 \, \log p},$$

uniformly for $\sigma \in (\frac{1}{2}, \frac{1}{2} + \varepsilon_0]$, since the rightmost double series converges. □

Lemma 24 (Arithmetic Carleson energy)**.** *Let*

$$U_{\det_2}(\sigma, t) \;:=\; \Re \log \det_2 \Big(I - A(\tfrac{1}{2} + \sigma + it)\Big) \;=\; -\sum_p \sum_{k \geq 2} \frac{p^{-k/2}}{k} \, e^{-k \log p \, \sigma} \, \cos(k \log p \, t), \qquad \sigma > 0,$$

where the series converges absolutely for every $\sigma > 0$. Then for every interval $I \subset \mathbb{R}$ with Carleson box $Q(I) := I \times (0, |I|]$,

$$\iint_{Q(I)} |\nabla U_{\det_2}|^2 \, \sigma \, dt \, d\sigma \;\leq\; \frac{|I|}{4} \sum_p \sum_{k \geq 2} \frac{p^{-k}}{k^2} \;=:\; K_0 \, |I|, \qquad K_0 := \frac{1}{4} \sum_p \sum_{k \geq 2} \frac{p^{-k}}{k^2} < \infty.$$

Proof. For a single mode $b \, e^{-\omega\sigma} \cos(\omega t)$ one has $|\nabla|^2 = b^2 \omega^2 e^{-2\omega\sigma}$, hence

$$\int_0^{|I|} \int_I |\nabla|^2 \, \sigma \, dt \, d\sigma \;\leq\; |I| \cdot \sup_{\omega > 0} \int_0^{|I|} \sigma \, \omega^2 e^{-2\omega\sigma} d\sigma \cdot b^2 \;\leq\; \tfrac{1}{4} \, |I| \, b^2.$$

With $b = p^{-k/2}/k$ and $\omega = k \log p$, summing over (p, k) gives the claim and the finiteness of K_0. □

Whitney scale and short–interval zero counts. Throughout the boundary-certificate route we work on Whitney boxes based at height T with

$$L = L(T) := \min\Big\{\frac{c}{\log\langle T\rangle},\ L_\star\Big\}, \qquad \langle T\rangle := \sqrt{1+T^2}, \qquad c \in (0,1] \text{ fixed.}$$

The only input about the *number* of zeros used below is the short-interval consequence of Riemann–von Mangoldt: there exist absolute constants $A_0, A_1 > 0$ such that for $T \geq 2$ and $0 < H \leq 1$,

$$N(T;H) \ := \ \#\{\rho = eta + i\gamma :\ \gamma \in [T, T+H]\} \ \leq \ A_0 \ + \ A_1\, H\, \log\langle T\rangle.$$

Lemma 25 (Annular Poisson–balayage L^2 bound)**.** *Let $I = [T-L, T+L]$, $Q_\alpha(I) = I \times (0, \alpha L]$, and fix $k \geq 1$. For $\mathcal{A}_k := \{\rho = eta + i\gamma :\ 2^k L < |T-\gamma| \leq 2^{k+1}L\}$ set*

$$V_k(\sigma, t) := \sum_{\rho \in \mathcal{A}_k} \frac{\sigma}{(t-\gamma)^2 + \sigma^2}.$$

Then

$$\iint_{Q_\alpha(I)} V_k(\sigma,t)^2\, \sigma\, dt\, d\sigma \ \ll_\alpha \ |I|\, 4^{-k}\, \nu_k,$$

where $\nu_k := \#\mathcal{A}_k$, and the implicit constant depends only on α.

Proof. Write $K_\sigma(x) := \sigma/(x^2+\sigma^2)$ and $V_k = \sum_{\rho\in\mathcal{A}_k} K_\sigma(\cdot - \gamma)$. Integrate over $t \in I$ first. For the diagonal terms, using $|t-\gamma| \geq 2^k L - L \geq 2^{k-1}L$ for $t \in I$ and $k \geq 1$,

$$\int_I K_\sigma(t-\gamma)^2\, dt = \sigma^2 \int_I \frac{dt}{ig((t-\gamma)^2 + \sigma^2 ig)^2} \ \leq \ \frac{L}{(2^{k-1}L)^2}\, \sigma.$$

Multiplying by the area weight σ and integrating $\sigma \in (0, \alpha L]$ gives a contribution $\ll_\alpha |I|\, 4^{-k}$ per γ, hence $\ll_\alpha |I|\, 4^{-k}\nu_k$ after summing. For off-diagonal terms, for $i \neq j$ one has on I that $K_\sigma(t-\gamma_j) \leq \sigma/(2^{k-1}L)^2$, hence

$$\int_I K_\sigma(t-\gamma_i) K_\sigma(t-\gamma_j)\, dt \ \leq \ \frac{\sigma}{(2^{k-1}L)^2} \int_{\mathbb{R}} K_\sigma(t-\gamma_i)\, dt = \frac{\pi\sigma}{(2^{k-1}L)^2},$$

and integrating $\sigma \in (0, \alpha L]$ with the extra factor σ yields $\ll_\alpha |I|\, 4^{-k}$. Summing over pairs (i,j) via a Schur test gives the stated bound (absorbing constants into $\ll_\alpha$). □

A.4. **Quantitative phase–velocity identity.** This subsection derives the quantitative identity linking the distribution $-w'$ to the off-axis zero data, in a form usable under Whitney localization.

Lemma 26 (Distributional phase–velocity identity for outer data)**.** *Let $U^* \in L^1_{\mathrm{loc}}(\mathbb{R})$ and define $w := \mathcal{H}[U^*] \in \mathcal{D}'(\mathbb{R})$ by the duality*

$$-\langle w, \varphi'\rangle \ = \ \int_{\mathbb{R}} U^*(t)\, (\mathcal{H}\varphi)'(t)\, dt \qquad \forall\, \varphi \in C_c^\infty(\mathbb{R}).$$

Then $w' = \mathcal{H}[(U^*)']$ in $\mathcal{D}'(\mathbb{R})$.

Proof. Let η_ε be a mollifier and set $U^*_\varepsilon := U^* * \eta_\varepsilon \in C^\infty(\mathbb{R})$. For smooth data one has $(\mathcal{H}[U^*_\varepsilon])' = \mathcal{H}[(U^*_\varepsilon)']$ pointwise. Since $U^*_\varepsilon \to U^*$ in L^1_{loc}, we have $(U^*_\varepsilon)' \to (U^*)'$ in $\mathcal{D}'$, and the Hilbert transform is continuous on $\mathcal{D}'$. Passing to the limit yields $w' = \mathcal{H}[(U^*)']$ in $\mathcal{D}'$. □

Lemma 27 (Local L^1 control for $\log|\xi|$ along vertical approach)**.** *Fix a compact interval $I \Subset \mathbb{R}$. Then the family $t \mapsto \log|\xi(\frac{1}{2}+\varepsilon+it)|$ is bounded in $L^1(I)$ uniformly for $\varepsilon \in (0,1]$. Moreover, for $\varepsilon, \varepsilon' \downarrow 0$ the difference $\log|\xi(\frac{1}{2}+\varepsilon+it)| - \log|\xi(\frac{1}{2}+\varepsilon'+it)|$ tends to 0 in $L^1(I)$.*

Proof. Write ξ in Hadamard form $\xi(s) = e^{a+bs}\prod_\rho igl(1-\frac{s}{\rho}igr)e^{s/\rho}$, where the product runs over nontrivial zeros ρ of ζ. Fix $I = [T_0, T_1] \Subset \mathbb{R}$ and $\varepsilon \in (0,1]$. Split the zeros into a finite set $\mathcal{Z}_R := \{\rho : \ |\Im\rho| \le R\}$ and the complement, with $R \ge 2 + \max(|T_0|, |T_1|)$. For $\rho \in \mathcal{Z}_R$, the map $t \mapsto \log|(\frac{1}{2}+\varepsilon+it)-\rho|$ lies in $L^1(I)$, with an $L^1(I)$ bound depending only on I and $\mathcal{Z}_R$ (local integrability of $\log|t-\gamma|$ near $\gamma = \Im\rho$). For $\rho \notin \mathcal{Z}_R$ and $t \in I$, one has $|(\frac{1}{2}+\varepsilon+it)/\rho| \ll_I 1/|\rho|$, so

$$\log\Big|\Big(1-\frac{\frac{1}{2}+\varepsilon+it}{\rho}\Big)e^{(\frac{1}{2}+\varepsilon+it)/\rho}\Big| = O_I igl(|\rho|^{-2}igr),$$

uniformly in $t \in I$ and $\varepsilon \in (0,1]$. Since $\sum_\rho |\rho|^{-2} < \infty$ (order 1 entire function), the tail contributes an absolutely convergent $L^\infty(I)$ error uniformly in ε. Combining these bounds gives $\sup_{\varepsilon\in(0,1]} \|\log|\xi(\frac{1}{2}+\varepsilon+i\cdot)|\|_{L^1(I)} < \infty$.

For the Cauchy property, write the difference as a sum over the same factorization. The finite set $\mathcal{Z}_R$ contributes a term that tends to 0 in $L^1(I)$ as $\varepsilon, \varepsilon' \downarrow 0$ by dominated convergence away from the finitely many points $t = \Im\rho$ and the local integrability of $\log|t-\Im\rho|$. The tail is uniformly $O_I\Big(\sum_{\rho\notin\mathcal{Z}_R}|\rho|^{-2}\Big)$ and hence uniformly small; letting $R \to \infty$ yields the $L^1(I)$-Cauchy claim. □

Lemma 28 (Local bounded-type control for $\mathcal{J}$)**.** *Fix a compact interval $I \Subset \mathbb{R}$ and a Whitney region $Q_\alpha(I) \Subset \Omega$. Assume that $\det_2(I - A(s))$ is holomorphic and nonvanishing on a neighborhood of $Q_\alpha(I)$, and that the arithmetic Carleson energy bound of Lemma 24 holds on $Q_\alpha(I)$. Then the ratio $\mathcal{J} = \det_2(I-A)/(\mathcal{O}\,\xi)$ constructed in Theorem 29 belongs to the Nevanlinna class (bounded type) on $Q_\alpha(I)$.*

Proof. Let $D := Q_\alpha(I)$ and write $\sigma = \Re s - \frac{1}{2}$. On D (simply connected), the hypotheses ensure that $\det_2(I-A)$ is holomorphic and nonvanishing, so $U_{\det_2} := \Re\log\det_2(I-A)$ is harmonic on D. The Carleson energy bound of Lemma 24 implies that $|\nabla U_{\det_2}|^2\,\sigma\,dt\,d\sigma$ is a Carleson measure on D. By the Fefferman–Stein/Carleson characterization of BMO boundary traces for harmonic functions on the half-plane (see Stein [10, Ch. IV] or Garnett [11, Ch. VI]), $U_{\det_2}$ admits nontangential boundary traces in $\mathrm{BMO}(I) \subset L^1(I)$. Hence $\det_2(I-A)$ is of bounded type on D.

Next, ξ is entire, so it is holomorphic on a neighborhood of $\overline{D}$ and therefore bounded on D; in particular $\xi \in H^\infty(D)$ and is of bounded type on D (and $1/\xi$ is meromorphic of bounded type on D).

Finally, the normalizing outer function $\mathcal{O}$ from Theorem 29 is holomorphic and zero-free on Ω, hence of bounded type on D. Since the Nevanlinna class is closed under products and quotients (where defined), it follows that $\mathcal{J} = \det_2(I-A)/(\mathcal{O}\,\xi)$ is of bounded type on D. □

Theorem 29 (Quantified phase–velocity identity and boundary passage)**.** *Let*

$$u_\varepsilon(t) := \log|\det_2(I - A(\tfrac{1}{2}+\varepsilon+it))| - \log|\xi(\tfrac{1}{2}+\varepsilon+it)|.$$

Then u_ε is uniformly L^1-bounded and Cauchy on every compact $I \Subset \mathbb{R}$ as $\varepsilon \downarrow 0$, hence $u_\varepsilon \to u$ in $L^1_{\mathrm{loc}}(\mathbb{R})$. Let $\mathcal{O}$ be the outer function on Ω with boundary modulus e^u and normalization $\mathcal{O}(\frac{3}{2}) > 0$, and set

$$\mathcal{J}(s) := \frac{\det_2(I - A(s))}{\mathcal{O}(s)\,\xi(s)}.$$

Then $|\mathcal{J}(\frac{1}{2}+it)| = 1$ for a.e. $t \in \mathbb{R}$. By Lemma 28, $\mathcal{J}$ is of bounded type on every Whitney region $Q_\alpha(I)$. Let $w \in \mathcal{D}'(\mathbb{R})$ denote the distributional boundary phase of $\mathcal{J}$ (defined modulo an additive constant). Then, for every compact interval $I \Subset \mathbb{R}$ and every nonnegative $\phi \in C_c^\infty(I)$,

$$\int_I \phi(t)\,(-w'(t))\,dt \;=\; \pi\int_I \phi(t)\,d\mu_{\mathrm{off}}(t) \;+\; \pi\int_I \phi(t)\,d\nu_{\mathrm{sing}}(t) \;+\; \pi\sum_{\gamma\in I} m_\gamma\,\phi(\gamma), \tag{7}$$

where:

- *μ_{off} is the Poisson balayage of the off–critical zeros $\rho = eta + i\gamma$ of ζ with $eta > \frac{1}{2}$, counted with multiplicity m_ρ;*
- *ν_{sing} is the (possibly zero) singular boundary measure of any singular inner factor in the canonical factorization of $\mathcal{J}$ on Ω; and*
- *the discrete sum ranges over boundary zeros/poles on $\Re s = \frac{1}{2}$, written $s = \frac{1}{2} + i\gamma$, with multiplicities m_γ.*

Proof. The L^1_{loc} convergence $u_\varepsilon \to u$ is as stated. The outer function $\mathcal{O}$ exists by Lemma 21.

Under the bounded-type hypothesis, $\mathcal{J}$ admits the canonical half-plane factorization into a unimodular constant, a Blaschke product over zeros in Ω, a (possibly trivial) singular inner factor, and an outer factor. Since $|\mathcal{J}(\frac{1}{2} + it)| = 1$ a.e., the outer factor is unimodular constant. Taking the distributional boundary argument w and differentiating in $\mathcal{D}'$, each factor contributes additively: the Blaschke product yields the Poisson balayage measure μ_{off}, the singular inner factor yields ν_{sing}, and boundary zeros/poles yield atomic Dirac masses. This is the phase-derivative computation for bounded-type functions on a half-plane; see, e.g., Garnett *Bounded Analytic Functions*, Ch. II, or Koosis *The Logarithmic Integral*, Vol. I. □

A.5. **Final assembly ingredients for the boundary wedge certificate (P+).** This subsection collects the three final analytic ingredients—Poisson plateau, Whitney box pairing, and Carleson-energy control—used to conclude (P+).

Poisson plateau lower bound. We prove the key lower bound for the Poisson kernel averaged over admissible windows, which converts discrete off-axis contributions into a uniform wedge aperture.

Lemma 30 (Poisson plateau lower bound)**.** *Let $\psi \in C_c^\infty(\mathbb{R})$ be even with $\psi \equiv 1$ on $[-1, 1]$ and* $\operatorname{supp}\psi \subset [-2, 2]$. *Then*

$$c_0(\psi) \;:=\; \inf_{0<b\le 1,\ |x|\le 1} (P_b * \psi)(x) \;\ge\; \frac{1}{2\pi}\arctan 2 \;>\; 0.$$

Proof. Since $\psi \ge \mathbf{1}_{[-1,1]}$, it suffices to compute $(P_b * \mathbf{1}_{[-1,1]})(x)$. For $|x| \le 1$,

$$(P_b * \mathbf{1}_{[-1,1]})(x) = \frac{1}{\pi}\int_{-1}^{1} \frac{b}{b^2 + (x-y)^2}\,dy = \frac{1}{2\pi}\Big(\arctan\frac{1-x}{b} + \arctan\frac{1+x}{b}\Big).$$

This expression is minimized over $0 < b \le 1$, $|x| \le 1$, at $(x, b) = (1, 1)$, yielding $\frac{1}{2\pi}\arctan 2$. □

From phase–velocity and CR–Green to (P+). We combine the phase–velocity identity with a Cauchy–Riemann/Green pairing on Whitney boxes to obtain the windowed phase control that underlies (P+).

Carleson energy bound for the logarithmic derivative. We now pass to the load-bearing box-energy control used in the wedge assembly. The key device is the *analytic inner reciprocal* $\mathcal{I} := B^2/\mathcal{J}_{\text{out}}$, which converts the poles of $\mathcal{J}_{\text{out}}$ (at ζ-zeros) into harmless zeros, yielding an honest inner function and a non-circular source of positivity for the Whitney gradient estimate.

Lemma 31 (Inner reciprocal and nonnegative potential)**.** *Let $\mathcal{J}_{\text{out}}$ be as in* (4) *and $B(s) := (s-1)/s$. Define*

$$\mathcal{I}(s) \;:=\; \frac{B(s)^2}{\mathcal{J}_{\text{out}}(s)} \;=\; \frac{B(s)\,\mathcal{O}_\zeta(s)\,\zeta(s)}{\det_2(I - A(s))}\,.$$

Then:

(1) *$\mathcal{I}$ is **holomorphic** on Ω. (The simple pole of ζ at $s = 1$ is canceled by B; zeros of ζ become zeros of $\mathcal{I}$; the denominator $\det_2(I - A)$ is nonvanishing on Ω.)*
(2) *$|\mathcal{I}(\frac{1}{2} + it)| = 1$ for Lebesgue-a.e. t. (On $\partial\Omega$: $|B| = 1$ and $|\mathcal{J}_{\text{out}}| = 1$ a.e.)*
(3) *$|\mathcal{I}(s)| \le 1$ for all $s \in \Omega$. (Phragmén–Lindelöf: $\log|\mathcal{I}|$ is subharmonic on Ω with boundary trace 0 a.e. and at most polynomial growth; see below.)*

In particular, the function

$$W(s) \;:=\; -\log|\mathcal{I}(s)| \;\geq\; 0 \qquad (s\in\Omega)$$

is nonnegative, and one has the identity

$$U(s) \;:=\; \log|\mathcal{J}_{\mathrm{out}}(s)| \;=\; 2\log|B(s)| \;+\; W(s) \qquad (s\in\Omega\setminus Z(\zeta)).$$

Proof. Part (1). Write $\mathcal{I} = B\,\mathcal{O}_\zeta\,\zeta/\det_2(I-A)$. The factor $B\zeta = (s-1)\zeta(s)/s$ is holomorphic on Ω (the simple pole of ζ at $s=1$ is canceled by the zero of $s-1$, and $s=0\notin\Omega$). The remaining factors $\mathcal{O}_\zeta$ (outer, zero-free) and $1/\det_2(I-A)$ (nonvanishing by (1)) are holomorphic on Ω. Hence $\mathcal{I}$ is holomorphic on Ω, with zeros exactly at the nontrivial zeros of ζ in Ω (same multiplicities).

Part (2). On $\partial\Omega$: $|B(\frac12+it)|^2 = |(-\frac12+it)/(\frac12+it)|^2 = (\frac14+t^2)/(\frac14+t^2) = 1$, and $|\mathcal{J}_{\mathrm{out}}(\frac12+it)| = 1$ a.e. by construction. Hence $|\mathcal{I}(\frac12+it)| = |B|^2/|\mathcal{J}_{\mathrm{out}}| = 1$ a.e.

Part (3). Since $\mathcal{I}$ is holomorphic on Ω, $u := \log|\mathcal{I}|$ is subharmonic on Ω. By Part (2), $u^*(\frac12+it) = 0$ for a.e. t. Moreover, $|\mathcal{I}(s)| \leq C(1+|t|)^N$ for some N and all $s = \frac12+\sigma+it$ with $\sigma\in(0,1]$ (this follows from the convexity bound for ζ, the absolutely convergent product for $\det_2$, and the Poisson-controlled modulus of $\mathcal{O}_\zeta$). Hence $u(s) = O(\log(2+|s|)) = o(|s|)$ as $|s|\to\infty$ in Ω. By the Phragmén–Lindelöf principle for the half-plane (e.g. Koosis, *The Logarithmic Integral*, Vol. I, Ch. III; or Garnett [11, Ch. I]): a subharmonic function on Ω with boundary trace ≤ 0 a.e. and growth $o(|s|)$ satisfies $u\leq 0$ on Ω. Hence $|\mathcal{I}|\leq 1$ and $W = -\log|\mathcal{I}| \geq 0$. □

Lemma 32 (Green potential representation for $U = \Re\log\mathcal{J}_{\mathrm{out}}$)**.** *Let $\mathcal{J}_{\mathrm{out}}$ be the outer-normalized unimodular ratio from* (4)*, and set*

$$U(s) \;:=\; \Re\log\mathcal{J}_{\mathrm{out}}(s) \;=\; \log|\mathcal{J}_{\mathrm{out}}(s)| \qquad (s\in\Omega\setminus Z(\zeta)).$$

For $a\in\Omega$, write $a^{\#} := 1-\overline{a}$ for reflection across the boundary line $\Re s = \frac12$, and define the half-plane Green kernel

$$G_\Omega(s,a) \;:=\; \log\left|\frac{s-a^{\#}}{s-a}\right|.$$

Let μ_ξ denote the ξ-zero counting measure in Ω,

$$\mu_\xi \;:=\; \sum_{\rho\in Z(\xi)\cap\Omega} m_\rho\,\delta_\rho,$$

where m_ρ is the multiplicity. Then, in the sense of distributions on Ω,

$$-\Delta\Big(U(s) \;-\; 2\log|B(s)|\Big) \;=\; 2\pi\,\mu_\xi, \qquad B(s) := \frac{s-1}{s}.$$

In particular, up to the explicit compensator $2\log|B|$, the field $U = \Re\log\mathcal{J}_{\mathrm{out}}$ is the Green potential generated by the ξ-zeros in Ω.

Proof. Since $\det_2(I-A)$ and $\mathcal{O}_\zeta$ are holomorphic and nonvanishing on Ω, the poles of $\mathcal{J}_{\mathrm{out}}$ in Ω coincide with the zeros of ζ in Ω, hence with the zeros of ξ in Ω. Moreover, $\mathcal{J}_{\mathrm{out}}$ has a zero at $s=1$ of order 2: one order comes from the simple pole of ζ at $s=1$ (so $1/\zeta$ has a simple zero) and one order comes from the factor $B(s) = (s-1)/s$.

We use the standard divisor identity for a meromorphic function F on a planar domain: $\Delta\log|F| = 2\pi\sum_{\mathrm{zeros}} m\,\delta - 2\pi\sum_{\mathrm{poles}} m\,\delta$ in distributions. Applying this to $\mathcal{J}_{\mathrm{out}}$ gives

$$-\Delta U \;=\; 2\pi\,\mu_\xi \;-\; 4\pi\,\delta_1.$$

Since $B(s)$ is holomorphic on Ω with a simple zero at $s=1$ and no poles in Ω, one has $\Delta\log|B| = 2\pi\,\delta_1$, hence $-\Delta(2\log|B|) = -4\pi\,\delta_1$. Subtracting yields the displayed identity. □

Proposition 33 (Height-dependent box-energy bound on Whitney scales)**.** *Let $U = \log|\mathcal{J}_{\text{out}}|$ on $\Omega \setminus Z(\zeta)$, and let $W = -\log|\mathcal{I}| \geq 0$ be the nonnegative potential from Lemma 31, so that $U = 2\log|B| + W$. For each Whitney interval $I = [t_0 - L, t_0 + L]$ with $L = c/\log\langle t_0\rangle$ and aperture $\alpha' > 1$, define the box energy*

$$E(I) \;:=\; \iint_{Q(\alpha' I)} |\nabla U|^2 \,\sigma\, dt\, d\sigma.$$

Then

$$E(I) \;\leq\; C_{\text{box}}(\alpha', c)\,|I|, \tag{8}$$

where $C_{\text{box}}(\alpha', c) < \infty$ is a ***height-independent*** *constant depending only on α' and the Whitney parameter c.*

In particular, the windowed-phase product satisfies

$$\sqrt{E(I)} \cdot L \;\leq\; \sqrt{C_{\text{box}}}\, L^{3/2} \;=\; \sqrt{C_{\text{box}}}\, \frac{c^{3/2}}{\log^{3/2}\langle t_0\rangle}\,, \tag{9}$$

which tends to 0 as $c \to 0$, uniformly in t_0.

Proof. Fix a Whitney interval $I = [t_0 - L, t_0 + L]$ with $L = c/\log\langle t_0\rangle$ and $\alpha' > 1$. Choose a slightly larger aperture parameter $\alpha'' > 2\alpha'$, and let $D := Q(\alpha'' I)$ (a dilated Whitney box).

Since $U = 2\log|B| + W$ and $B = (s-1)/s$ is explicit and smooth on D (for t_0 large, the clip $L \leq L_\star$ keeps D away from $s = 1$), $\nabla(2\log|B|)$ contributes $O_{\alpha'}(|I|)$ to the weighted energy. It therefore suffices to bound the W-energy:

$$E_W(I) \;:=\; \iint_{Q(\alpha' I)} |\nabla W|^2 \,\sigma\, dt\, d\sigma.$$

Step 1 (Whitney neutralization of $\mathcal{I}$). By Lemma 31, $\mathcal{I}$ is an inner function on Ω with zeros exactly at the nontrivial zeros of ζ in Ω. Factor $\mathcal{I} = e^{i\theta} B_{\text{near}} B_{\text{far}} S$, where B_{near} is the finite Blaschke product over zeros $\rho = \beta + i\gamma$ of $\mathcal{I}$ with $|\gamma - t_0| \leq \alpha'' L$, B_{far} is the Blaschke product of the remaining zeros, and S is the (possibly trivial) singular inner factor. By the Riemann–von Mangoldt short-interval bound, B_{near} has at most $A_0 + A_1\alpha'' c = O_{\alpha'',c}(1)$ factors.

Define the neutralized field

$$\widetilde{W}(s) \;:=\; W(s) + \log|B_{\text{near}}(s)| \;=\; -\log|B_{\text{far}}(s)| \;-\; \log|S(s)|.$$

Every term on the right is ≥ 0 (each inner factor has modulus ≤ 1 on Ω), so $\widetilde{W} \geq 0$ on Ω. On $\partial\Omega$ ($\sigma = 0$): all inner factors have boundary modulus 1, so $\widetilde{W} = 0$. Moreover, $\widetilde{W}$ is *harmonic* on D: the zeros in B_{far} have $|\gamma - t_0| > \alpha'' L$, hence lie outside the t-span of D, and S is zero-free.

The $O(1)$ Blaschke factors in B_{near} each contribute $O_{\alpha'}(|I|)$ to the weighted Dirichlet energy on $Q(\alpha' I)$, giving

$$E_{\text{near}} \;:=\; \iint_{Q(\alpha' I)} |\nabla \log|B_{\text{near}}||^2\, \sigma \;=\; O_{\alpha',c}(|I|).$$

Step 2 (Carleson-measure energy bound for $\widetilde{W}$). Rather than bounding the *pointwise* sup of $\widetilde{W}$ on ∂D and applying a gradient estimate, we bound the *weighted Dirichlet energy* of $\widetilde{W}$ directly using the Carleson measure property of inner functions.

Key fact (Carleson measure theorem for inner functions; see Garnett [11, Ch. IX] or Cima–Matheson–Ross, *The Cauchy Transform*, Ch. 6). Let g be an inner function on Ω with canonical factorization $g = e^{i\theta} B_g S_g$. Then the measure $|g'/g|^2\, \sigma\, d\sigma\, dt$ is a **Carleson measure** on Ω, and for every interval $J \subset \mathbb{R}$:

$$\iint_{Q(J)} |g'/g|^2\, \sigma\, d\sigma\, dt \;\leq\; C_{\text{Carl}}\,(\#\{\text{zeros of } g \text{ with ordinate in } 3J\} \;+\; \nu_{S_g}(3J)), \tag{10}$$

where ν_{S_g} is the singular measure of S_g, $3J$ is a fixed dilate of J, and C_{Carl} is a universal constant.

Now, the neutralized field $\widetilde{W} = -\log|B_{\text{far}}\, S|$ satisfies $|\nabla\widetilde{W}|^2 = |(B_{\text{far}}S)'/(B_{\text{far}}S)|^2$ on D (where $B_{\text{far}}S$ is zero-free and hence $\widetilde{W}$ is harmonic). Apply (10) with $g = B_{\text{far}}\, S$ and J a constant dilate of I:

$$\iint_{Q(\alpha' I)} |\nabla\widetilde{W}|^2\,\sigma \;\le\; C_{\text{Carl}}\,\big(N_{\text{dilate}} \;+\; \nu_S(3\alpha' I)\big), \tag{11}$$

where $N_{\text{dilate}} := \#\{\text{zeros of } \mathcal{I} \text{ with ordinate in } 3\alpha' I \text{ but not in } \alpha'' I\}$ is the number of *far zeros near the dilated interval*. By the Riemann–von Mangoldt bound on Whitney scale: $N_{\text{dilate}} \le A_0 + A_1 \cdot 3\alpha'|I| \cdot \log\langle t_0\rangle = A_0 + 6A_1\alpha' c = O_{\alpha',c}(1)$.

For the singular inner mass $\nu_S(3\alpha' I)$: since S is part of the inner function $\mathcal{I}$ from Lemma 31 and $|\mathcal{I}| \le 1$, the singular measure satisfies the same Carleson-type bound as the zero-counting measure (see Garnett [11, Ch. IX, Thm. 1.1]). On Whitney scale, $\nu_S(3\alpha' I) = O_{\alpha',c}(1)$.

Therefore (11) gives

$$\iint_{Q(\alpha' I)} |\nabla\widetilde{W}|^2\,\sigma \;\le\; C_{\text{Carl}} \cdot O_{\alpha',c}(1) \;=:\; C_4(\alpha', c).$$

This is a **height-independent constant**.

Step 3 (assembly). Since $W = \widetilde{W} - \log|B_{\text{near}}|$: $|\nabla W|^2 \le 2|\nabla\widetilde{W}|^2 + 2|\nabla\log|B_{\text{near}}||^2$. And $U = 2\log|B| + W$: $|\nabla U|^2 \le 2|\nabla(2\log|B|)|^2 + 2|\nabla W|^2$. Combining:

$$E(I) \;\le\; 4C_4(\alpha', c) \;+\; O_{\alpha',c}(|I|) \;\le\; C_{\text{box}}(\alpha', c)\,|I|,$$

where $C_{\text{box}}(\alpha', c)$ is a **finite constant independent of the height** t_0. This is (8). For the product (9): $\sqrt{E(I)} \le \sqrt{C_{\text{box}}\,|I|} = \sqrt{2C_{\text{box}}L}$, hence

$$\sqrt{E(I)} \cdot L \;\le\; \sqrt{2C_{\text{box}}}\,L^{3/2} \;=\; \sqrt{2C_{\text{box}}}\,\frac{c^{3/2}}{\log^{3/2}\langle t_0\rangle} \;\to\; 0 \quad \text{as } c \to 0, \text{ uniformly in } t_0.$$

□

A.6. **From windowed phase control to the wedge.** We convert the established windowed phase bounds into an almost-everywhere wedge inclusion for the boundary values of $\mathcal{J}_{\text{out}}$.

Definition 34 (Admissible window class with atom avoidance)**.** Fix an even C^∞ window ψ with $\psi \equiv 1$ on $[-1, 1]$ and $\operatorname{supp}\psi \subset [-2, 2]$. For an interval $I = [t_0 - L, t_0 + L]$, an aperture $\alpha' > 1$, and a parameter $\varepsilon \in (0, \frac{1}{4}]$, define $\mathcal{W}_{\text{adm}}(I; \varepsilon)$ to be the set of C^∞, nonnegative, mass-1 bumps ϕ supported in the fixed dilate $2I = [t_0 - 2L, t_0 + 2L]$ that can be written as

$$\phi(t) \;=\; \frac{1}{Z}\frac{1}{L}\,\psi\!\left(\frac{t - t_0}{L}\right) m(t), \qquad Z = \int_{2I} \frac{1}{L}\psi\!\left(\frac{t - t_0}{L}\right) m(t)\,dt,$$

where $2I := [t_0 - 2L, t_0 + 2L]$ and the mask $m \in C^\infty(2I; [0, 1])$ satisfies:

(i) *Atom avoidance.* There is a union of disjoint open subintervals $E = igcup_{j=1}^{J} J_j \subset I$ with total length $|E| \le \varepsilon L$ such that $m \equiv 0$ on E and $m \equiv 1$ on $I \setminus E'$, where each transition layer $E' \setminus E$ has thickness $\le \varepsilon L$.

(ii) *Uniform smoothness.* $\|m'\|_\infty \lesssim (\varepsilon L)^{-1}$ and $\|m''\|_\infty \lesssim (\varepsilon L)^{-2}$ with implicit constants independent of I, t_0, L and of the number/placement of the holes $\{J_j\}$.

Every $\phi \in \mathcal{W}_{\text{adm}}(I; \varepsilon)$ is supported in $2I$. This class contains the unmasked profile $\varphi_{L,t_0}(t) = Z_0^{-1}L^{-1}\psi((t - t_0)/L)$ with $Z_0 := \int_{-2}^{2}\psi(x)\,dx$ (take $E = \varnothing$, $m \equiv 1$) and also allows dodging boundary atoms by punching out small neighborhoods while keeping total deleted length $\le \varepsilon L$.

Lemma 35 (Uniform Poisson–energy bound for admissible tests)**.** *Let V_ϕ be the Poisson extension of $\phi \in \mathcal{W}_{\mathrm{adm}}(I;\varepsilon)$ to the half-plane, and fix a cutoff to $Q(\alpha' I)$ with $\alpha' > 1$ as in the CR–Green pairing. Then there exists a finite constant $\mathcal{A}_{\mathrm{adm}}(\psi,\varepsilon,\alpha') < \infty$, depending only on $(\psi,\varepsilon,\alpha')$, such that*

$$\iint_{Q(\alpha' I)} |\nabla V_\phi(\sigma,t)|^2\, \sigma\, dt\, d\sigma \;\le\; \mathcal{A}_{\mathrm{adm}}(\psi,\varepsilon,\alpha')^2\, L.$$

Proof. Let $\phi(t) = Z^{-1}L^{-1}\psi((t-t_0)/L)m(t)$ be an admissible test. By scaling of the Poisson kernel and the uniform bounds on m, m', m'' from Definition 34, the H^1-size of ϕ (equivalently the $L^2(\sigma)$ Dirichlet energy of its Poisson extension on a fixed aperture box) is controlled uniformly by a constant depending only on $(\psi,\varepsilon,\alpha')$, times $L^{1/2}$. Squaring yields the stated $\lesssim L$ energy bound with $\mathcal{A}_{\mathrm{adm}}(\psi,\varepsilon,\alpha')$. □

Lemma 36 (Cutoff pairing on boxes)**.** *Fix parameters $\alpha' > \alpha > 1$. Let $\chi_{L,t_0} \in C_c^\infty(\mathbb{R}^2_+)$ satisfy $\chi \equiv 1$ on $Q(\alpha I)$, $\operatorname{supp}\chi \subset Q(\alpha' I)$, $\|\nabla\chi\|_\infty \lesssim L^{-1}$ and $\|\nabla^2\chi\|_\infty \lesssim L^{-2}$. Let V_ϕ be the Poisson extension of $\phi \in \mathcal{W}_{\mathrm{adm}}(I;\varepsilon)$. Then one has the Green pairing identity*

$$\int_{\mathbb{R}} u(t)\,\phi(t)\,dt \;=\; \iint_{Q(\alpha' I)} \nabla U\cdot\nabla(\chi_{L,t_0}\,V_\phi)\,dt\,d\sigma \;+\; \mathcal{R}_{\mathrm{side}} \;+\; \mathcal{R}_{\mathrm{top}},$$

with remainders satisfying

$$|\mathcal{R}_{\mathrm{side}}| + |\mathcal{R}_{\mathrm{top}}| \;\lesssim\; \Big(\iint_{Q(\alpha' I)} |\nabla U|^2\,\sigma\Big)^{1/2}\cdot\Big(\iint_{Q(\alpha' I)} ig(|\nabla\chi|^2\,|V_\phi|^2 + |\nabla V_\phi|^2 ig)\,\sigma\Big)^{1/2}.$$

Proof. Let $Q := Q(\alpha' I)$. Assume U is C^2 on $\overline{Q}$ and harmonic on Q, with boundary trace $u(t) = U(0,t)$ on the bottom edge $\{\sigma = 0\}$. Since $\chi_{L,t_0}V_\phi$ is compactly supported in $\overline{Q}$ and smooth on Q, Green's identity gives

$$\iint_Q \nabla U\cdot\nabla(\chi V_\phi)\,dt\,d\sigma \;=\; \int_{\partial Q}(\chi V_\phi)\,\partial_n U\,ds \;-\; \iint_Q (\chi V_\phi)\,\Delta U\,dt\,d\sigma.$$

Since $\Delta U = 0$ on Q, only the boundary integral remains. On the bottom edge one has $\partial_n = -\partial_\sigma$, $\chi \equiv 1$, and $V_\phi(0,t) = \phi(t)$, hence that contribution equals

$$\int_I \phi(t)\,(-\partial_\sigma U)(0,t)\,dt.$$

A.7. **Whitney box pairing and local oscillation functional.** We organize the remaining bookkeeping: pairing Whitney boxes across scales and bounding the oscillation functional needed to pass from local control to the global wedge statement.

If U is the real part of a holomorphic logarithm $U = \Re\log J$ with $|J(\frac{1}{2}+it)| = 1$ a.e., then $U(0,t) = 0$ a.e. and $-\partial_\sigma U(0,t) = \partial_t \operatorname{Arg} J(\frac{1}{2}+it)$ in distributions by Cauchy–Riemann; in particular, this term is the tested boundary phase derivative in Lemma 37 below. The remaining boundary pieces (two vertical sides and the top edge) are, by definition, the remainders $\mathcal{R}_{\mathrm{side}} + \mathcal{R}_{\mathrm{top}}$.

For the remainder estimate, we apply Cauchy–Schwarz in the scale-invariant measure $\sigma\,dt\,d\sigma$ on Q:

$$ig|\mathcal{R}_{\mathrm{side}}ig| + ig|\mathcal{R}_{\mathrm{top}}ig| \;\lesssim\; \Big(\iint_Q |\nabla U|^2\,\sigma\Big)^{1/2}\Big(\iint_Q ig|\nabla(\chi V_\phi)|^2\,\sigma\Big)^{1/2}.$$

Expanding $\nabla(\chi V_\phi) = \chi\,\nabla V_\phi + (\nabla\chi)\,V_\phi$ yields

$$\iint_Q ig|\nabla(\chi V_\phi)|^2\,\sigma \;\lesssim\; \iint_Q ig(|\nabla V_\phi|^2 + |\nabla\chi|^2|V_\phi|^2 ig)\,\sigma,$$

which gives the displayed estimate. □

Lemma 37 (CR–Green pairing for boundary phase)**.** *Let J be analytic on Ω with a.e. boundary modulus $|J(\frac{1}{2}+it)|=1$, and write $\log J = U + iW$ on Ω, so U is harmonic with $U(\frac{1}{2}+it)=0$ a.e. Fix a Whitney interval $I=[t_0-L,t_0+L]$ and let V_ϕ be the Poisson extension of $\phi \in \mathcal{W}_{\mathrm{adm}}(I;\varepsilon)$. Then, with a cutoff χ_{L,t_0} as in Lemma 36,*

$$\int_{\mathbb{R}} \phi(t)\, ig(-W'(t))\, dt \;=\; \iint_{Q(\alpha' I)} \nabla U \cdot \nabla(\chi_{L,t_0} V_\phi)\, dt\, d\sigma \;+\; \mathcal{R}_{\mathrm{side}} \;+\; \mathcal{R}_{\mathrm{top}},$$

and the remainders satisfy the same estimate as in Lemma 36. In particular, by Cauchy–Schwarz and Lemma 35, there is a constant $C_{\mathrm{rem}}(\alpha',\psi)$ such that

$$\int_{\mathbb{R}} \phi(t)\, ig(-w'(t))\, dt \;\le\; C_{\mathrm{rem}}(\alpha',\psi)\Big(\iint_{Q(\alpha' I)} |\nabla U|^2\, \sigma\Big)^{1/2}.$$

Proof. On the bottom edge $\{\sigma=0\}$ the outward normal is $\partial_n = -\partial_\sigma$. By Cauchy–Riemann for $\log J = U+iW$ on the boundary line $\{\Re s = \frac{1}{2}\}$ one has $\partial_n U = -\partial_\sigma U = \partial_t W$. Thus the bottom-edge term in Green's identity is

$$-\int_{\partial Q\cap\{\sigma=0\}} \chi\, V_\phi\, \partial_n U\, dt = -\int_{\mathbb{R}} \phi(t)\, \partial_t W(t)\, dt = \int_{\mathbb{R}} \phi(t)\, ig(-w'(t))\, dt,$$

which yields the stated identity after including the interior term and remainders. The final inequality is Cauchy–Schwarz together with the uniform Poisson-energy bound from Lemma 35. □

Proposition 38 (Length-independent upper bound for admissible tests)**.** *Let J be holomorphic on $\Omega\setminus Z(\zeta)$ with a.e. boundary modulus 1, write $\log J = U+iW$ on $\Omega\setminus Z(\zeta)$, and let $-w'$ denote the boundary phase distribution. For every interval $I=[t_0-L,t_0+L]$, every $\phi\in\mathcal{W}_{\mathrm{adm}}(I;\varepsilon)$, and every fixed cutoff to $Q(\alpha' I)$,*

$$\int_{\mathbb{R}} \phi(t)\,(-w')(t)\, dt \;\le\; C_{\mathrm{test}}(\psi,\varepsilon,\alpha')\Big(\iint_{Q(\alpha' I)} |\nabla U|^2\, \sigma\, dt\, d\sigma\Big)^{1/2} \tag{12}$$

with $C_{\mathrm{test}}(\psi,\varepsilon,\alpha') := C_{\mathrm{rem}}(\alpha',\psi)\,\mathcal{A}_{\mathrm{adm}}(\psi,\varepsilon,\alpha')$ independent of I, t_0, L.

Proof. Apply Lemma 37 with $\phi\in\mathcal{W}_{\mathrm{adm}}(I;\varepsilon)$ and absorb the window-side constants into $C_{\mathrm{test}}(\psi,\varepsilon,\alpha')$. □

Lemma 39 (Whitney–uniform wedge bound)**.** *Fix parameters $\alpha'>1$ and $\varepsilon\in(0,\frac{1}{4}]$. Fix the Whitney schedule: set*

$$L(t_0) \;:=\; \frac{c}{\log\langle t_0\rangle}, \qquad c\in(0,1].$$

Then for every Whitney interval $I=[t_0-L,t_0+L]$ and the corresponding cutoff $\psi_{L,t_0}(t) := \psi((t-t_0)/L) = Z_0 L\,\varphi_{L,t_0}(t)$ (so $\psi_{L,t_0}\equiv 1$ on I), the windowed phase satisfies the uniform bound

$$\int_{\mathbb{R}} \psi_{L,t_0}(t)\,(-w'(t))\, dt \;\le\; Z_0\, C_{\mathrm{test}}(\psi,\varepsilon,\alpha')\sqrt{2C_{\mathrm{box}}(\alpha',c)}\; L^{3/2} \;\le\; Z_0\, C_{\mathrm{test}}\sqrt{2C_{\mathrm{box}}}\,\frac{c^{3/2}}{\log^{3/2}2} \;=:\; \pi\,\Upsilon_{\mathrm{Whit}}(c).$$

Choosing $c>0$ sufficiently small makes $\Upsilon_{\mathrm{Whit}}(c)<\frac{1}{2}$, bounding the windowed phase variation on every Whitney interval. (A separate local-to-global argument—not supplied in this version—would be needed to promote this windowed bound to the full boundary wedge (P+)*.)*

Proof. Since $\psi_{L,t_0} = Z_0 L\,\varphi_{L,t_0}$, Proposition 38 gives

$$\int \psi_{L,t_0}\,(-w') \;=\; Z_0 L\int \varphi_{L,t_0}\,(-w') \;\le\; Z_0 L\cdot C_{\mathrm{test}}\sqrt{E(I)}\,.$$

By the bound (9) of Proposition 33, $\sqrt{E(I)}\cdot L \le \sqrt{2C_{\mathrm{box}}}\,L^{3/2} = \sqrt{2C_{\mathrm{box}}}\,c^{3/2}/\log^{3/2}\langle t_0\rangle$, so

$$Z_0 L\cdot C_{\mathrm{test}}\sqrt{E(I)} \;=\; Z_0 C_{\mathrm{test}}\cdot\sqrt{E(I)}\cdot L \;\le\; Z_0 C_{\mathrm{test}}\sqrt{2C_{\mathrm{box}}}\,\frac{c^{3/2}}{\log^{3/2}2}\,,$$

which is uniformly bounded in t_0 and tends to 0 as $c\to 0$. □

Remark 40 (Energy-budget mechanism underlying (P+)). The mechanism that drives (P+) admits a transparent decomposition into "source cost" and "field cost" (cf. the explicit-formula duality between primes and zeros).

Source (prime) side. The regularized determinant $\det_2(I - A(s))$ is holomorphic and *zero-free* on Ω (Lemma 2); it contributes the arithmetic Carleson constant $K_0 = \frac{1}{4}\sum_p \sum_{k\geq 2} p^{-k}/k^2 < \infty$ (Lemma 24). This constant is unconditionally finite because the prime series converges on the half-plane—it encodes the total "cost" of the irreducible multiplicative generators.

Field (zero) side. The zeros of ζ in Ω contribute to $|\nabla U|^2$ through the Green-potential representation (Lemma 32). After neutralizing the $O(1)$ near-field zeros by a local Blaschke product, the harmonic residual $\widetilde{U}$ satisfies a boundary bound $|\widetilde{U}| \leq M = O(\log\langle T\rangle)$ on the dilated Whitney box, by the Poisson-kernel averaging of the far-field Green contributions together with the short-interval Riemann–von Mangoldt density estimate. The interior gradient estimate for harmonic functions then gives a box energy $E(I) \leq C\log^2\langle T\rangle \cdot |I|$ that may grow with height but is *overwhelmed* by the Whitney scaling $L^{3/2} = (c/\log\langle T\rangle)^{3/2}$.

The decisive algebraic fact is that the product $\sqrt{E(I)} \cdot L = O(c^{3/2}/\log^{3/2}\langle T\rangle)$ tends to 0 as $c \to 0$ *uniformly in height*. The Whitney–uniform wedge (Lemma 39) converts this vanishing product into a phase bound $\Upsilon_{\mathrm{Whit}}(c) < \frac{1}{2}$, and the local-to-global lemma delivers (P+). In short: bounded near-field cost + Poisson-averaged far-field control $\Rightarrow$ height-dependent box energy beaten by Whitney shrinkage $\Rightarrow$ small windowed phase $\Rightarrow$ boundary wedge.

Remark 41 (Status of the (P+) route). The boundary wedge (P+) has *not* been established unconditionally in this version. Lemma 39 bounds the windowed phase variation on each Whitney interval by $\pi\Upsilon_{\mathrm{Whit}}(c)$, which can be made $< \pi/2$ by choosing c small. However, a separate *local-to-global* argument (promoting windowed control to the full Lebesgue-a.e. wedge inclusion) is required but not supplied here. Moreover, the (P+) statement as written (a single global wedge for all heights) would, combined with the Smirnov transport in §4, rule out zeros for *all* $\Re s > 1/2$—which is far stronger than the claimed $\Re s \geq 0.6$ result and is not what the current analysis establishes.

The direct-contradiction proof of Theorem 1 in §4.1 is the intended primary route. Using the Carleson measure theorem for inner functions (Proposition 33), the energy bound is now *height-independent*: $E(I) \leq C_{\mathrm{box}}|I|$. The direct contradiction closes with a fixed choice of c.

Appendix B. Supplementary computational verification protocol (verifier commands and expected fields)

There are two verification modes:

- **Verification using shipped artifacts:** verify the shipped JSON artifacts match Table 1.
- **Optional regeneration check (supplementary):** rerun the verifier to regenerate the artifacts from scratch.

Prerequisites. From the bundle's `compute/` directory, install the ARB/ball-arithmetic bindings:

```
pip install -r requirements.txt
```

Verification using shipped artifacts: check shipped JSON artifacts.

- **Rectangle artifact** `artifacts/theta_certify_sigma06_07_t0_20_outer_zeta_proj.json`. Check (at minimum):
  - `results.ok = true`
  - `results.theta_hi = 0.9999928763...  < 1`
  - `results.processed_boxes = 380764`
- **Pick artifact** `artifacts/pick_sigma0599_raw_zeta_N16.json`. Check (at minimum):
  - `pick.delta_cert = 0.594...`
  - `pick.P_spd_at_0 = true`

  - `pick.tail_l1_partial_hi` (diagnostic L1 tail sum)
- **Pick artifact** `artifacts/pick_sigma06_raw_zeta_N16.json`. Check (at minimum):
  - `pick.delta_cert = 0.594...`
  - `pick.P_spd_at_0 = true`
  - `pick.tail_l1_partial_hi` (diagnostic L1 tail sum)
- **Pick artifact** `artifacts/pick_sigma07_raw_zeta_N16.json`. Check (at minimum):
  - `pick.delta_cert = 0.627...`
  - `pick.P_spd_at_0 = true`
  - `pick.tail_l1_partial_hi` (diagnostic L1 tail sum)

Optional regeneration check (supplementary): exact command lines. Run the verifier from the bundle's `compute/` directory (or the corresponding directory in a repository checkout). The following commands reproduce the primary artifacts (line breaks are for readability):
1) Rectangle certification (`theta_certify`).

```
python verify_attachment_arb.py \
  --theta-certify \
  --arith-gauge outer_zeta_proj \
  --arith-P-cut 2000 \
  --rect-sigma-min 0.6 --rect-sigma-max 0.7 \
  --rect-t-min 0.0 --rect-t-max 20.0 \
  --outer-mode midpoint \
  --outer-P-cut 2000 \
  --outer-T 50.0 --outer-n 2001 \
  --theta-init-n-sigma 10 --theta-init-n-t 50 \
  --theta-min-sigma-width 0.0001 --theta-min-t-width 0.001 \
  --theta-max-boxes 500000 \
  --prec 260 \
  --theta-out artifacts/theta_certify_sigma06_07_t0_20_outer_zeta_proj.json \
  --progress
```

2) Pick certification at $\sigma_0 = 0.599$ (`pick_certify`).

```
python verify_attachment_arb.py \
  --pick-certify \
  --pick-sigma0 0.599 \
  --pick-N 16 \
  --pick-coeff-count 128 \
  --pick-K 512 \
  --pick-rho 0.4 \
  --pick-rho-bound 0.5 \
  --arith-gauge raw_zeta \
  --arith-P-cut 2000 \
  --prec 1024 \
  --pick-out artifacts/pick_sigma0599_raw_zeta_N16.json
```

3) Pick certification at $\sigma_0 = 0.6$ (`pick_certify`).

```
python verify_attachment_arb.py \
  --pick-certify \
  --pick-sigma0 0.6 \
  --pick-N 16 \
  --pick-coeff-count 128 \
  --pick-K 512 \
```

```
  --pick-rho 0.4 \
  --pick-rho-bound 0.5 \
  --arith-gauge raw_zeta \
  --arith-P-cut 2000 \
  --prec 1024 \
  --pick-out artifacts/pick_sigma06_raw_zeta_N16.json
```

4) Pick certification at $\sigma_0 = 0.7$ (`pick_certify`).

```
python verify_attachment_arb.py \
  --pick-certify \
  --pick-sigma0 0.7 \
  --pick-N 16 \
  --pick-coeff-count 128 \
  --pick-K 512 \
  --pick-rho 0.4 \
  --pick-rho-bound 0.5 \
  --arith-gauge raw_zeta \
  --arith-P-cut 2000 \
  --outer-mode rigorous \
  --outer-P-cut 2000 \
  --prec 1024 \
  --pick-out artifacts/pick_sigma07_raw_zeta_N16.json
```

Meaning of successful verification. The verifier uses *ball arithmetic*: each computed quantity is an interval enclosure (midpoint plus radius) and every operation propagates rounding error outward. Thus each check is a formal inequality of the form `upper bound < 1` or `directed-rounding` $\mathrm{LDL}^\top$ `succeeds with positive pivots`. If the verification checks above pass, then the numerical inequalities summarized in Table 1 are within the logic of ball arithmetic.

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Email address: `jon@recognitionphysics.org`
Email address: `arahnamab@gmail.com`